



TECHNISCHE
UNIVERSITÄT
DRESDEN

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Project work

TRANSIENT B-SPLINE MPM INSIDE THE FEAP FRAMEWORK

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Institute for
Structural Analysis

Project work

Transient B-spline MPM inside the FEAP framework

Transient B-spline MPM inside
the FEAP framework

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Submitted on 30th of February 2345



Conceptual Formulation of the Project Work

PW06/25

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Field of Study: ACCESS Master Course

Topic: **Transient B-spline MPM inside the FEAP framework**
Transiente B-spline MPM im FEM-Programm FEAP

Objective:

The Material Point Method is a particle-based simulation technique, which solves partial differential equations by projecting the particles onto a background grid. The most basic form utilizes Lagrangian shape functions for these projections, which gives rise to the so-called cell-crossing error. As it is caused by the jump in the shape functions' gradient, improved results can be achieved by using higher order B-spline shape functions.

The finite element framework FEAP was recently augmented a basic MPM toolkit, allowing for any element leveraging the standard FEAP quadrature routines to be used with the new method. Currently, this is limited to standard linear shape functions, but as FEAP also contains an IGA module, an extension to B-spline shape functions is feasible.

Additionally, the current implementation is limited to quasi-static simulations. To improve the versatility of the framework, support for transient simulations is desirable.

To improve the accuracy of the MPM simulations, the task of the project work is to efficiently implement transient B-spline MPM for FEAP, utilizing as many of the available components as possible. The components of the project work are as follows:

- implementation of B-spline MPM using the FEAP IGA module
- support for transient MPM simulations in FEAP
- mesh-based generation of the initial configuration of the material points
- representative benchmarks, comparing the developed B-spline MPM to linear MPM and FEM

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Abstract

The Finite Element Analysis Program (FEAP) framework contains a preliminary Material Point Method (MPM) module. However, its reliance on quasi-static simulations and linear shape functions causes severe cell-crossing errors due to discontinuous shape function gradients. To improve numerical accuracy and versatility, this thesis efficiently implements a transient B-spline MPM formulation within FEAP. First, a mesh-based particle generation approach is developed to accurately establish the initial material point configuration. By leveraging the built-in Isogeometric Analysis (IGA) module of FEAP, higher-order continuous B-spline shape functions are adopted to produce smooth strain fields, fundamentally eliminating stress errors. Furthermore, the framework is extended to enable full support for transient MPM and Isogeometric Analysis Material Point Method (IGA-MPM) simulations through implicit NEWMARK time integration and a hybrid Particle-in-Cell(PIC)/Fluid-Implicit-Particle(FLIP) kinematic advection scheme. Representative benchmarks verify the implementation, demonstrating that the developed B-spline MPM significantly outperforms conventional linear MPM in suppressing non-physical stress oscillations.

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1 Introduction

The numerical simulation of structural dynamics and solid mechanics relies on spatial and temporal discretization techniques [4, 19, 32, 33]. The Finite Element Method (FEM) serves as a standard method in computational mechanics for solving partial differential equations governing continuum bodies [4]. The finite element framework Finite Element Analysis Program (FEAP) provides a computational environment for such analyses [31]. Using a mesh-based approach, the FEM maps structural geometries and tracks history-dependent material variables. However, the FEM exhibits limitations in problems involving large deformations, fragmentation, or penetration [5, 11, 18]. Under these conditions, the LAGRANGIAN mesh experiences distortion and element entanglement. To address these limitations, hybrid methods were introduced to combine EULERIAN and LAGRANGIAN descriptions. Among these, the Material Point Method (MPM) is a particle-based technique that solves partial differential equations by projecting LAGRANGIAN material points onto a fixed EULERIAN background grid [12].

FEAP incorporated a basic MPM module to address large deformation problems. The current implementation is restricted to linear shape functions and static simulations, lacking the capability to generate initial material point configurations for curved geometries. Furthermore, utilizing C^0 -continuous shape functions causes the cell-crossing error [29]. Caused by the discontinuity in the shape functions' gradient, material points traversing element boundaries induce non-physical stress oscillations [29]. A solution to mitigate this error is to enhance the spatial continuity of the background grid by integrating higher-order basis functions, such as Non-Uniform Rational B-Splines (NURBS) or B-splines [15, 25]. The availability of an Isogeometric Analysis (IGA) module within FEAP [30] enables the extension of the MPM architecture towards B-spline shape functions, which defines the IGA-MPM [21]. Therefore, to improve the versatility and accuracy of MPM simulations, the objective of this project work is to implement a transient B-spline MPM formulation within FEAP. To address the aforementioned issues, the core components involve implementing the B-spline MPM via the FEAP IGA module, extending the framework for transient simulations, developing a mesh-based initialization approach for material points, and evaluating the method through benchmarks to compare the B-spline MPM against the linear MPM and the FEM.

To validate the proposed framework and demonstrate its capability to overcome the aforementioned numerical issues, three representative benchmark tests are investigated. First, a static cantilever bar is analyzed to evaluate the spatial and quadrature convergence behavior. Subsequently, the transient dynamics are tested using a one-dimensional elastic bar subjected to a pulse impact. Finally, the simulation of the dynamic bending of an elastic cantilever beam is conducted to evaluate the stability of the stress field under large deformations.

This work is structured into five parts. Chapter 2 presents the theoretic fundamentals concerning continuum mechanics, B-splines, NURBS, and the hyperelastic material model. Chapter 3 outlines the discretization methods, transitioning from the strong to the weak form equations, and details the time integration and kinematic advection schemes. The formulations of implicit MPM, the mechanics of the cell-crossing error, and the theoretical

shift to IGA-MPM are presented in Chapter 4. The algorithmic design and implementation within the FEAP architecture are detailed in Chapter 5. The work concludes with Chapter 6, providing numerical examples and validation through static convergence and dynamic benchmarks.

2 Fundamentals

This chapter gives an overview of the basics of continuum mechanics and an introduction to the utilized discretization techniques. It was generally compiled from [1, 7, 14, 28].

2.1 Kinematics

A continuum body with closed boundaries consists of an infinite number of material particles occupying a specific domain defined in a three-dimensional space. To discuss the motion of individual particles, a fixed rectangular cartesian coordinate system with base vectors $\mathbf{e}_i$ ($i = 1, 2, 3$) is introduced.

2.1.1 Body configurations

Assume that the body starts moving at time $t = 0$ until it occupies a new region at time t . The body at $t = 0$ is denoted as $\mathcal{B}_0 \subset \mathbb{R}^3$, occupying the region $\mathcal{R}_0$ in the three-dimensional Euclidean space. Furthermore, at time t , the body is indicated by $\mathcal{B} \subset \mathbb{R}^3$ and occupies the region $\mathcal{R}$. $\mathbf{X}$ and $\mathbf{x}$ are the position vectors of a particle in $\mathcal{B}_0$ and $\mathcal{B}$, respectively. The state of $\mathcal{B}_0$ is the initial configuration, and $\mathcal{B}$ is the current configuration. Figure 2.1.1 shows the body and quantities in both configurations. The motion of the body $\mathcal{B}$ is defined by the mapping φ such that

$$\mathbf{x} = \varphi(\mathbf{X}, t), \quad (2.1.1a)$$

$$\mathbf{X} = \varphi^{-1}(\mathbf{x}, t). \quad (2.1.1b)$$

and the displacements $\mathbf{u}$ are defined as the difference between the positions

$$\mathbf{u}(\mathbf{X}, t) = \mathbf{x}(\mathbf{X}, t) - \mathbf{X}. \quad (2.1.2)$$

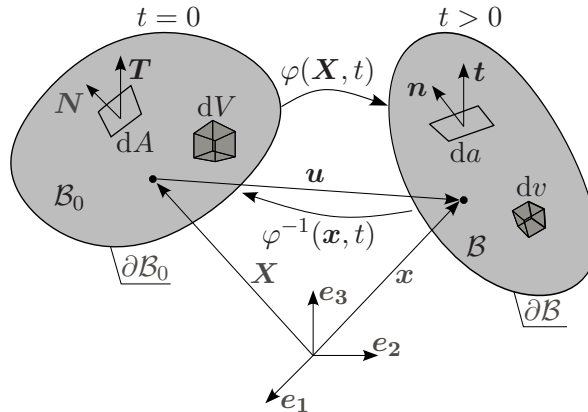


Figure 2.1.1: Initial and current configurations of a continuum body with associated quantities.

The deformation gradient $\mathbf{F}$ is defined by the position vector $\mathbf{X}$ and $\mathbf{x}$

$$\mathbf{F} = \frac{\partial \mathbf{x}}{\partial \mathbf{X}} = \nabla \varphi(\mathbf{X}, t) \quad (2.1.3)$$

Equation (2.1.2) together with Equation (2.1.3) leads to

$$\mathbf{F} = \frac{\partial \mathbf{x}}{\partial \mathbf{X}} = \mathbf{I} + \nabla \mathbf{u} \quad (2.1.4)$$

where $\mathbf{I}$ is the second order identity tensor. Using this, the relations between the infinitesimal measures can be written as

$$d\mathbf{x} = \mathbf{F}d\mathbf{X}, \quad (2.1.5a)$$

$$da = J\mathbf{F}^{-T}d\mathbf{A}, \quad (2.1.5b)$$

$$dv = JdV, \quad (2.1.5c)$$

where $d\mathbf{X}$, dA and dV are the infinitesimal line, area and volume elements in the initial configuration respectively, while $d\mathbf{x}$, da and dv are in the current configuration. J is the determinant of the deformation gradient $J = \det(\mathbf{F})$. The infinitesimal areas are defined as

$$d\mathbf{A} = \mathbf{N}dA, \quad (2.1.6a)$$

$$d\mathbf{a} = \mathbf{n}da, \quad (2.1.6b)$$

where $\mathbf{N}$ and $\mathbf{n}$ are the outward normal vectors in the initial configuration body surface $\partial\mathcal{B}_0$ and current configuration body surface $\partial\mathcal{B}$ respectively.

The velocity vector $\mathbf{v}$ is the rate of the particle displacement. As a result, $\mathbf{v}$ is calculated by deriving Equation (2.1.2) with respect to the time

$$\mathbf{v}(X, t) = \frac{\partial \mathbf{u}(X, t)}{\partial t} = \frac{\partial \mathbf{x}(\mathbf{X}, t)}{\partial t} - \frac{\partial \mathbf{X}}{\partial t}, \quad (2.1.7)$$

and because $\mathbf{X}$ is constant over time, Equation (2.1.7) becomes

$$\mathbf{v}(X, t) = \frac{\partial \mathbf{x}(\mathbf{X}, t)}{\partial t}. \quad (2.1.8)$$

Acceleration is defined as the rate of change of velocity with respect to time. Thus, the acceleration vector of a particle is deduced by

$$\mathbf{a}(\mathbf{X}, t) = \frac{\partial \mathbf{v}(\mathbf{X}, t)}{\partial t} = \frac{\partial^2 \mathbf{u}(\mathbf{X}, t)}{\partial t^2}, \quad (2.1.9)$$

The rate of the deformation gradient is given by

$$\dot{\mathbf{F}} = \frac{d}{dt} \frac{\partial \mathbf{x}}{\partial \mathbf{X}} = \frac{\partial}{\partial \mathbf{X}} \frac{d\mathbf{x}}{dt} = \frac{\partial}{\partial \mathbf{X}} \mathbf{v}(\mathbf{X}, t), \quad (2.1.10)$$

using the chain rule

$$\dot{\mathbf{F}} = \frac{\partial}{\partial \mathbf{X}} \mathbf{v}(\mathbf{X}, t) = \frac{\partial \mathbf{v}}{\partial \mathbf{x}} \frac{\partial \mathbf{x}}{\partial \mathbf{X}} = \mathbf{L}\mathbf{F}, \quad (2.1.11)$$

where $\mathbf{L}$ is the velocity gradient.

2.1.2 Strain measures

$\mathbf{F}$ is inadequate for measuring deformation because it changes during rotation, which violates the condition that the quantity should remain constant in the absence of any deformation. Thus, alternative measures are defined, starting with the right CAUCHY-GREEN tensor

$$\mathbf{C} = \mathbf{F}^T \mathbf{F}, \quad (2.1.12)$$

and the left CAUCHY-GREEN tensor

$$\mathbf{B} = \mathbf{F} \mathbf{F}^T. \quad (2.1.13)$$

$\mathbf{C}$ and $\mathbf{B}$ are constant for a rigid body motion, often utilized for strain energy. On the other hand, the GREEN-LAGRANGE strain tensor is zero in the case of rigid body motion

$$\mathbf{E} = \frac{1}{2}(\mathbf{C} - \mathbf{I}), \quad (2.1.14)$$

as well as the ALMANSI strain tensor

$$\mathbf{e} = \frac{1}{2}(\mathbf{I} - \mathbf{B}^{-1}), \quad (2.1.15)$$

where $\mathbf{I}$ is the second-order identity tensor. Furthermore, $\mathbf{F}$ can be described based on the polar decomposition

$$\mathbf{F} = \mathbf{R} \mathbf{U} = \mathbf{V} \mathbf{R}, \quad (2.1.16)$$

where $\mathbf{U}$ and $\mathbf{V}$ are the right and left stretch tensors, respectively,

$$\mathbf{U} = \sqrt{\mathbf{C}}, \quad (2.1.17a)$$

$$\mathbf{V} = \sqrt{\mathbf{B}}, \quad (2.1.17b)$$

and $\mathbf{R}$ is the orthogonal rotation tensor

$$\mathbf{R} = \mathbf{F} \mathbf{U}^{-1}. \quad (2.1.18)$$

2.2 Stress measures

Stress means the arrangement of force applied on a defined surface of a body. As a result, a force $d\mathbf{f}$ applied at a material particle is defined

$$d\mathbf{f} = \mathbf{t} da, \quad (2.2.1)$$

where Figure 2.2.1 defines the CAUCHY traction vector $\mathbf{t}$ as the force per unit area. Considering the equilibrium of an infinitesimal tetrahedron, the force $d\mathbf{f}$ on the base area da with normal $\mathbf{n}$ is countered by the forces $d\mathbf{f}_i$ acting on the coordinate faces (Figure 2.2.1a). These forces are expressed through traction vectors $\mathbf{t}_i$, which relate directly to the CAUCHY stress components σ_{ij} shown in Figure 2.2.1b.

The relationship between traction vector $\mathbf{t}$ and the CAUCHY stress tensor $\boldsymbol{\sigma}$ is established by the CAUCHY stress theorem

$$\mathbf{t} = \boldsymbol{\sigma} \mathbf{n}, \quad (2.2.2)$$

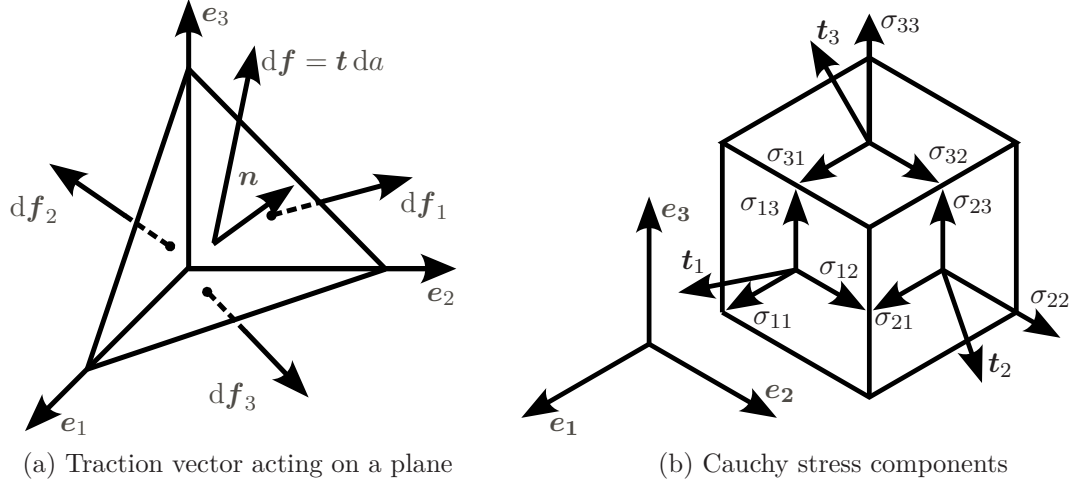


Figure 2.2.1: CAUCHY traction vector and stress tensor shown on infinitesimal elements.

in matrix form

$$\begin{Bmatrix} t_1 \\ t_2 \\ t_3 \end{Bmatrix} = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{bmatrix} \begin{Bmatrix} n_1 \\ n_2 \\ n_3 \end{Bmatrix}. \quad (2.2.3)$$

The CAUCHY stress tensor refers to true stress commonly. Another PIOLA-KIRCHHOFF stress tensor $\mathbf{S}$ conjugate to GREEN-LAGRANGE strain tensor $\mathbf{E}$, is often used in computations

$$\mathbf{S} = \mathbf{F}^{-1} \mathbf{J} \boldsymbol{\sigma} \mathbf{F}^{-T}. \quad (2.2.4)$$

2.3 Conservation laws

Conservation laws are the bedrock of classical physics, governing the evolution of fundamental quantities: mass, linear and angular momentum, and energy. These principles dictate that such quantities are preserved within a closed system, changing only through internal transformation or external exchange. The following sections provide an overview of these foundational balance equations.

2.3.1 Conservation of mass

The principle of mass conservation asserts that the total mass within an isolated system remains invariant over time. This global balance is expressed as

$$m = \int_{B_0} \rho_0 \, dV = \int_B \rho \, dv, \quad (2.3.1)$$

where m is the total mass of a body, and ρ_0 and ρ are the densities of a material point in the reference and current configurations, respectively. Equation (2.3.1) means that the mass does not change in time. Hence its time derivative yields zero

$$\frac{d}{dt} m = \frac{d}{dt} \int_{B_0} \rho_0 \, dV = \frac{d}{dt} \int_B \rho \, dv = 0. \quad (2.3.2)$$

The determinant of the deformation gradient relates ρ and ρ_0 derived by using Equation (2.1.5c) and Equation (2.3.1) as follows

$$\rho_0 = J\rho, \quad (2.3.3)$$

2.3.2 Balance of linear momentum

The balance of linear momentum states that the rate of change of a system's momentum must equate to the sum of all applied external forces. In the current configuration, the total linear momentum $\mathbf{l}$ is defined as

$$\mathbf{l} = \int_{\mathcal{B}} \rho \mathbf{v} \, dv, \quad (2.3.4)$$

and the total external force vector is

$$\mathbf{p} = \int_{\mathcal{B}} \rho \mathbf{b}_m \, dv + \int_{\partial \mathcal{B}} \mathbf{t} \, da, \quad (2.3.5)$$

where $\mathbf{b}_m$ is the vector of body forces per unit mass applied on the particles. Hence, the balance of linear momentum is expressed as

$$\frac{d}{dt} \int_{\mathcal{B}} \rho \mathbf{v} \, dv = \int_{\mathcal{B}} \rho \mathbf{b} \, dv + \int_{\partial \mathcal{B}} \rho \mathbf{t} \, da, \quad (2.3.6)$$

with $\mathbf{b} = \rho \mathbf{b}_m$. Using the mass conservation law, the CAUCHY stress theorem, and the GAUSS divergence theorem

$$\int_{\partial \mathcal{B}} \rho \mathbf{t} \, da = \int_{\mathcal{B}} \nabla \cdot \boldsymbol{\sigma} \, dv. \quad (2.3.7)$$

Equation (2.3.6) is expressed as

$$\rho \frac{d\mathbf{v}}{dt} = \rho \mathbf{b} + \nabla \cdot \boldsymbol{\sigma}, \quad (2.3.8)$$

which represents the local form of the balance of linear momentum.

2.3.3 Balance of Angular Momentum

The principle of angular momentum balance asserts that the material time rate of change of the total angular momentum is equal to the sum of the moments generated by external body forces and surface tractions, taken with respect to a fixed origin. This integral balance is formulated as

$$\frac{d}{dt} \int_{\mathcal{B}} \mathbf{x} \times \rho \mathbf{v} \, dv = \int_{\mathcal{B}} \mathbf{x} \times \rho \mathbf{b} \, dv + \int_{\partial \mathcal{B}} \mathbf{x} \times \rho \mathbf{t} \, da. \quad (2.3.9)$$

By applying mathematical transformations, it can be demonstrated that Equation (2.3.9) can only be satisfied if the CAUCHY stress tensor is symmetric

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}^T, \quad (2.3.10)$$

2.4 NURBS and IGA introduction

This section establishes the core framework of the geometric representations utilized in the numerical simulations of this thesis. Building upon the aforementioned continuum mechanics foundations, it bridges the gap between geometric description and numerical analysis. Conventional numerical methods, such as the FEM, often encounter difficulties when handling complex geometries due to the discrepancy between Computer-Aided Design (CAD) representations and generated computational meshes. To alleviate this, IGA, originally proposed by HUGHES et al. [15], employs the exact basis functions utilized in CAD for numerical discretization.

Serving as a methodological prelude, this section introduces the underlying concepts and mathematical formulations of the geometric entities implemented throughout this study. The fundamental premise of IGA is to utilize standard CAD representations, such as NURBS or T-splines, to model exact computational meshes as well as their subsequent kinematics and deformations. This unification of geometry and analysis offers superior refinement characteristics and high-order inter-element continuity.

The theoretical exposition commences with the definition of B-splines, which form the basis of curve construction in modern CAD systems. Subsequently, the formulation is extended to B-spline surfaces and solids to facilitate multi-dimensional modeling. Finally, NURBS are introduced as a generalized and versatile implementation capable of exactly representing conic sections.

Knot vectors

Knot vectors define the local parametric space of B-splines. Formally, a knot vector is a non-decreasing sequence of one-dimensional coordinates, denoted as $\Xi = \{\xi_1, \xi_2, \dots, \xi_{n+p+1}\}$, where p represents the polynomial degree and n corresponds to the number of basis functions.

Knot vectors critically govern the support locality and continuity of the basis functions. They are typically classified as uniform for equally spaced knots, or open (clamped) when the first and last knots are repeated $p + 1$ times. Open knot vectors force the resulting B-spline to interpolate the initial and terminal control points, thereby facilitating model construction and boundary condition enforcement.

Basis functions

Basis functions are the main component of B-splines. They are recursively defined using the Cox-de Boor recursion formula

$$N_{i,p}(\xi) = \frac{\xi - \xi_i}{\xi_{i+p} - \xi_i} N_{i,p-1}(\xi) + \frac{\xi_{i+p+1} - \xi}{\xi_{i+p+1} - \xi_{i+1}} N_{i+1,p-1}(\xi) \quad (2.4.1)$$

with the base case defined as piecewise constant

$$N_{i,0}(\xi) = \begin{cases} 1 & \text{if } \xi_i \leq \xi < \xi_{i+1}, \\ 0 & \text{otherwise.} \end{cases} \quad (2.4.2)$$

For $p = 0$, the basis functions are piecewise constant between ξ_i and ξ_{i+1} , and for $p = 1$, they are linear between ξ_i and ξ_{i+2} with a peak at ξ_{i+1} . While these match the constant

and linear basis functions used in FEM, higher-order (≥ 2) B-spline basis functions differ significantly, offering superior smoothness and continuity. A fundamental comparison is shown in Figure 2.4.1, contrasting the C^1 continuity of B-splines with the C^0 continuity of Lagrange polynomials.

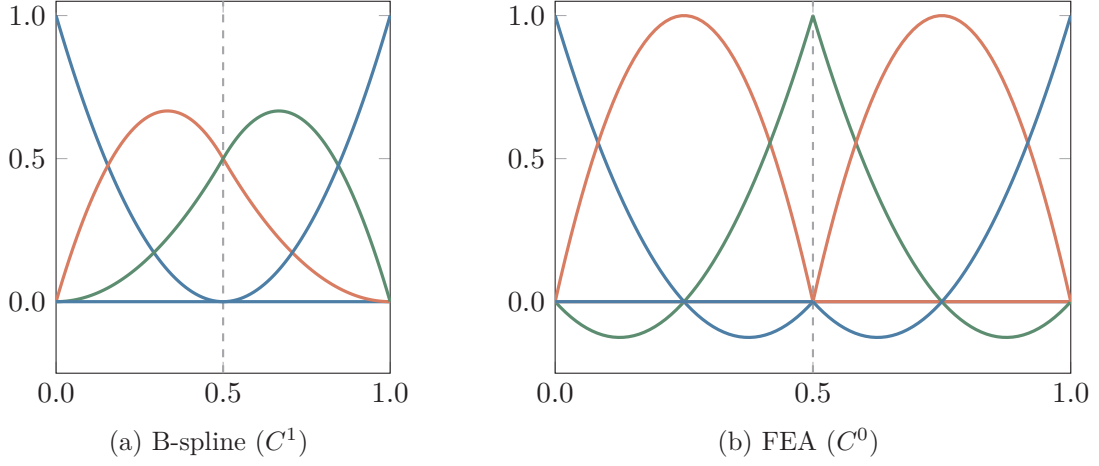


Figure 2.4.1: Comparison of B-spline and FEA basis functions ($p = 2$).

Furthermore, the number and shape of the basis functions heavily depend on the knot vector. In IGA, open knot vectors where the first and last knots are repeated $p + 1$ times are predominantly utilized. The primary reason for this choice is that open knot vectors yield basis functions that are interpolatory at the boundaries of the parametric domain. This crucial property facilitates the direct imposition of essential boundary conditions. Concurrently, these basis functions inherently preserve the fundamental property of a partition of unity, which is formulated as

$$\forall \xi \in [\xi_1, \xi_{n+p+1}] : \sum_{i=1}^n N_{i,p}(\xi) = 1. \quad (2.4.3)$$

This property ensures the exact reproduction of rigid-body motions by the basis functions, which is essential for numerical convergence. For closed B-splines utilizing unclamped knot vectors, an overlapping scheme is typically applied, equating the first p knots to the final p knots to preserve this property and close the domain. Additionally, the support of $N_{i,p}$ remains compact within $[\xi_i, \xi_{i+p+1}]$ and strictly positive. This distinctive attribute sets them apart from LAGRANGIAN basis functions, resulting in a strictly positive mass matrix.

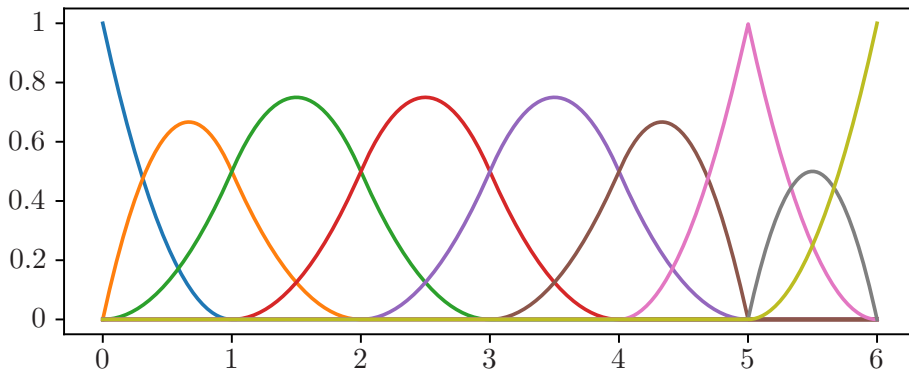


Figure 2.4.2: B-spline basis functions for $p = 2$ and $\Xi = [0, 0, 0, 1, 2, 3, 4, 5, 5, 6, 6, 6]$.

2.4.1 Geometric Entities in Isogeometric Analysis

Utilizing the basis functions described in the section 2.4, computational geometries can be constructed by introducing a set of control points. This direct link between mathematical discretization and geometric definition forms the cornerstone of the isogeometric method.

B-spline curves

Applying these basis functions to control points $\mathbf{B}_i$ and combining them leads to the definition of a B-spline curve in $\mathbb{R}^d$

$$\mathbf{C}(\xi) = \sum_{i=1}^n N_{i,p}(\xi) \mathbf{B}_i. \quad (2.4.4)$$

Unlike interpolatory Lagrangian polynomials, B-spline curves are generally non-interpolatory. As clearly illustrated in Figure 2.4.3, the curve itself only touches the control points at specific locations corresponding to knots that are repeated at least p times within the knot vector, such as the start and end of a clamped curve. For the interior domain, the curve forms a smooth, parabola-like shape for each control point, rounding out the sharp corners of the underlying control polygon.

Another fundamental property of B-splines is their intrinsic C^{p-1} continuity. This high level of smoothness is only disturbed at repeated interior knots, where the degree of continuity is reduced by one per repetition of the knot. This characteristic allows for the precise representation of complex geometries featuring both smooth transitions and sharp corners within a single parametric domain.

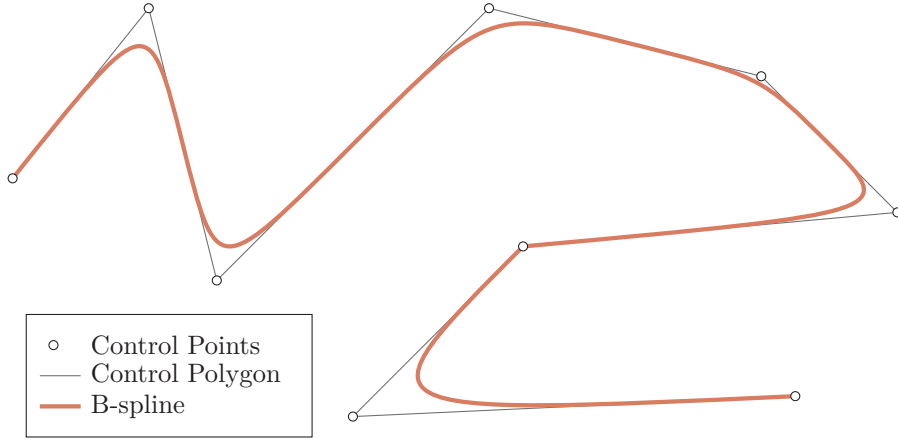


Figure 2.4.3: B-spline basis with the knot vector of Figure 2.4.2.

Similar to LAGRANGIAN basis functions, B-splines can undergo various refinement strategies to enhance numerical precision. Specifically, h -refinement introduces additional knots and control points, whereas p -refinement increases the basis function order while preserving the underlying geometry. Furthermore, k -refinement represents a distinct spline-specific approach that integrates both techniques to elevate the polynomial degree while preserving maximal continuity. The subsequent sections of this chapter elaborate on all of these methodologies.

***h*-refinement**

This method, alternatively termed knot insertion, parallels conventional finite element refinement by adapting the computational model directly. To achieve this, a new knot is embedded into the existing knot vector to yield a valid configuration. Consequently, this incorporation necessitates the generation of an additional control point. Preserving the original curve geometry and continuity profile then requires repositioning the existing control points relative to the newly inserted knot.

Assuming $\bar{\xi} \in [\xi_k, \xi_{k+1}]$, the new knot vector is $\Xi = \{\xi_1, \xi_2, \dots, \xi_k, \bar{\xi}, \xi_{k+1}, \dots, \xi_{n+p+1}\}$ leading to a new set of control points $\bar{\mathbf{B}}_{1..n+1}$ derived from the old points $\mathbf{B}_{1..n}$ by

$$\bar{\mathbf{B}}_i = \alpha_i \mathbf{B}_i + (1 - \alpha_i) \mathbf{B}_{i-1} \quad (2.4.5)$$

with

$$\alpha_i = \begin{cases} 1, & 1 \leq i \leq k - p \\ \frac{\bar{\xi} - \xi_i}{\xi_{i+p} - \xi_i}, & k - p + 1 \leq i \leq k \\ 0, & k + 1 \leq i \leq n + p + 2 \end{cases}. \quad (2.4.6)$$

This principle is illustrated in Figure 2.4.4 using a representative configuration governed by the knot vector $\Xi = \{0, 0, 0, 1, 1, 1\}$. Although the curve geometry remains invariant, both the quantity and spatial coordinates of the control points are modified, whereas the boundary points stay fixed due to the clamped knot vector constraints. Furthermore, the resulting alterations in the basis functions facilitate an enhanced representation of deformations, analogous to conventional *h*-refinement procedures in finite element analysis.

***p*-refinement**

This method operates analogously to its Finite Element Analysis (FEA) counterpart by elevating the polynomial degree of the B-spline basis functions, a process commonly designated as order elevation. To preserve underlying curve properties particularly concerning continuity and derivatives the multiplicity of each distinct knot must be concurrently increased. The resulting modified knot vector is subsequently utilized to define the refined B-spline. However, updating the control points presents greater complexity, leading to various implementation strategies. The primary approach entails decomposing the B-spline into individual BÉZIER segments, subjecting each segment to order elevation, and subsequently reassembling them into the global B-spline. A comprehensive algorithmic description and alternative methods exceed the scope of this contribution and can be found in references such as [25]. This transformation process is graphically demonstrated in Figure 2.4.5.

***k*-refinement**

Combining *h*- and *p*-refinement exhibits commutative failure, necessitating a strict sequence. As shown in Figure 2.4.6, inserting knots before order elevation yields larger knot vectors with higher multiplicities, whereas elevating the order first avoids this. Because order elevation doubles each unique knot while insertion adds one, the second strategy produces a shorter, highly continuous knot vector with identical parametrization, making it universally preferable. Reduced continuity can be subsequently achieved via knot insertion if needed.

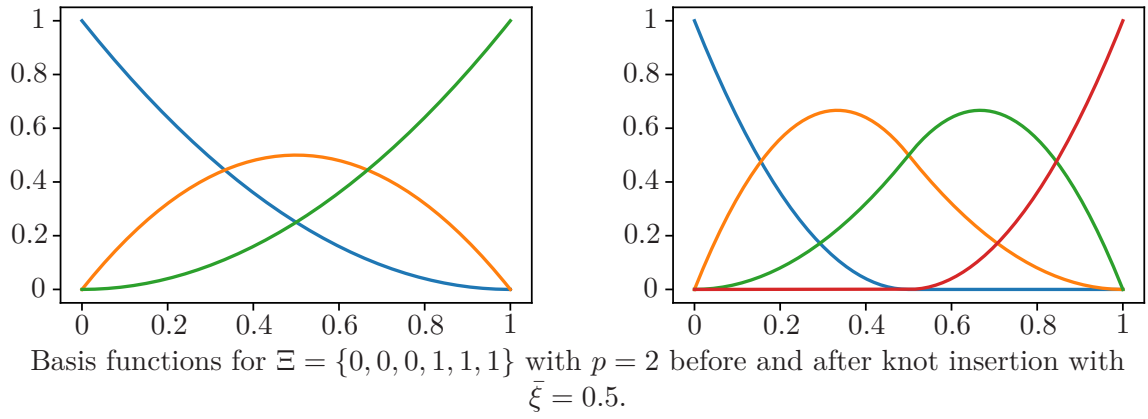
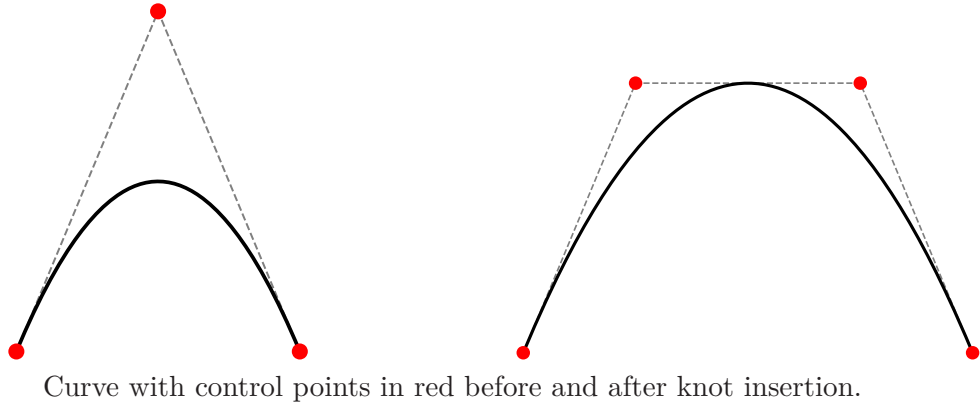


Figure 2.4.4: Visualization of h -refinement.

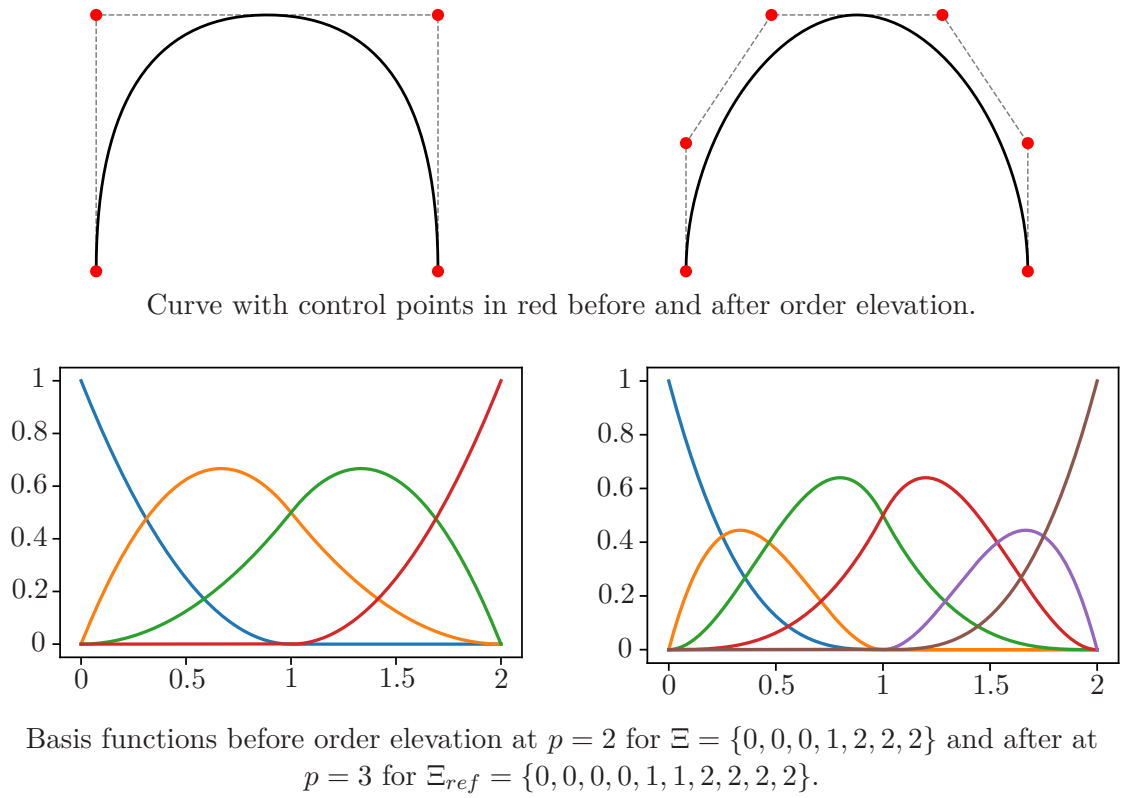


Figure 2.4.5: Visualization of p -refinement.

Consequently, in B-spline and NURBS design, the order should be defined prior to geometry refinement via additional knots or control points.

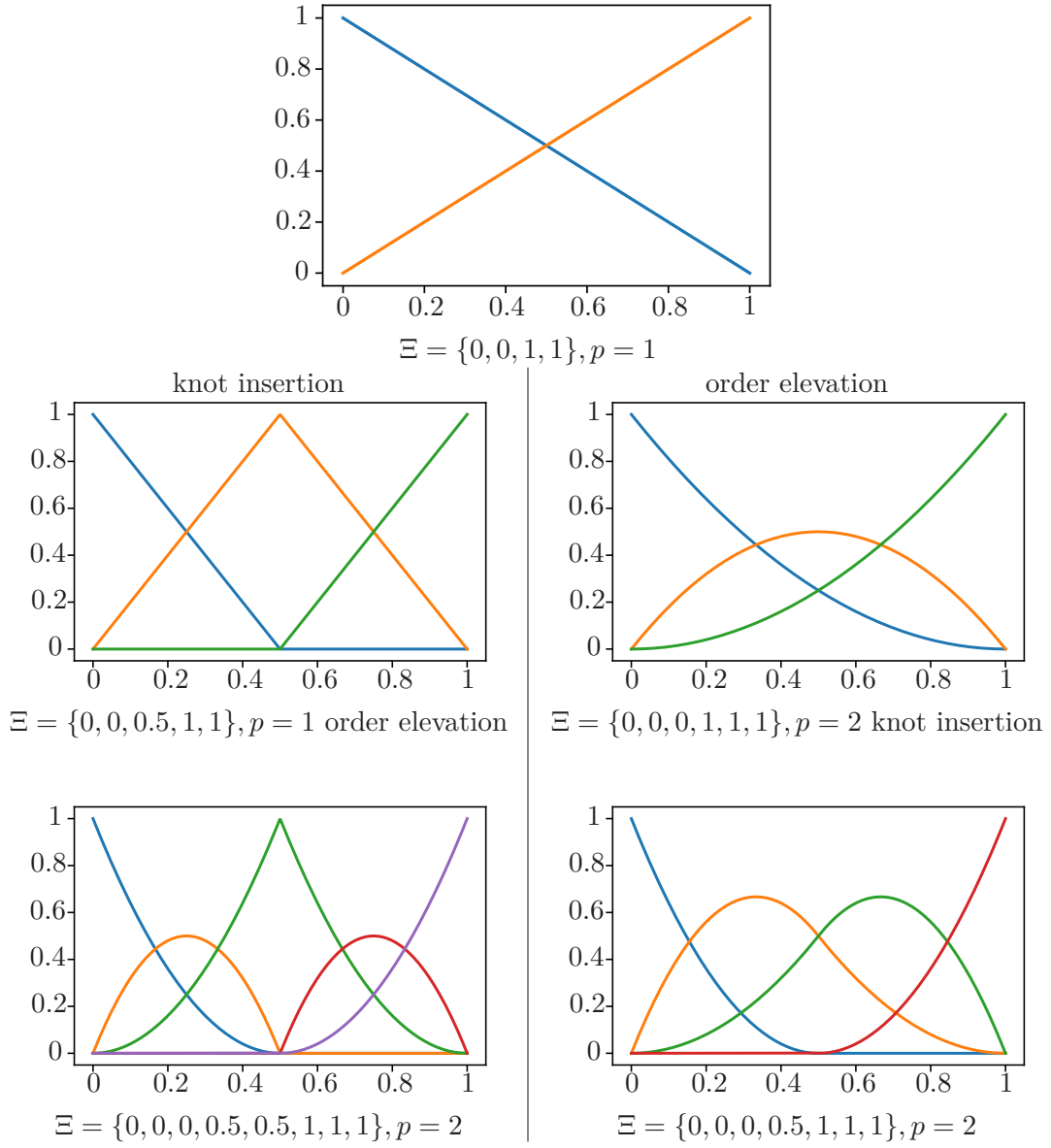


Figure 2.4.6: Visualization of combining refinement strategies. The left shows knot insertion before order elevation and the right the other way around.

Definition of surfaces and solids

A surface can be defined similar to a curve using two sets of basis functions $N_{i,p}(\xi)$ and $M_{j,q}(\eta)$

$$\mathcal{S}(\xi, \eta) = \sum_{i=1}^n \sum_{j=1}^m N_{i,p}(\xi) M_{j,q}(\eta) \mathbf{B}_{i,j}, \quad (2.4.7)$$

with $\mathbf{B}_{i,j}$ as a set of control points. Similar, adding a third basis functions $L_{k,r}(\zeta)$ creates a B-spline solid

$$\mathbf{S}(\xi, \eta, \zeta) = \sum_{i=1}^n \sum_{j=1}^m \sum_{k=1}^l N_{i,p}(\xi) M_{j,q}(\eta) L_{k,r}(\zeta) \mathbf{B}_{i,j,k}. \quad (2.4.8)$$

Defining the parametric space requires two or three knot vectors for surfaces and solids, respectively. In IGA, an element corresponds to the parametric knot span $[\xi_k, \xi_{k+1}] \times [\eta_j, \eta_{j+1}] \times [\zeta_k, \zeta_{k+1}]$. Such patches or multiple patches joined by shared control points form a model. Since patch boundaries exhibit only C^0 -continuity, minimizing the patch count is highly advisable.

Extension to NURBS

NURBS generalize B-splines by weighting each basis function with w_i , yielding the rational basis functions

$$R_{i,p}(\xi) = \frac{w_i N_{i,p}(\xi)}{\sum_{j=1}^n w_j N_{j,p}(\xi)}. \quad (2.4.9)$$

The curve is defined as

$$\mathbf{C}(\xi) = \sum_{i=1}^n R_{i,p}(\xi) \mathbf{B}_i, \quad (2.4.10)$$

with the control points $\mathbf{B}$. Geometrically, weights $\mathbf{B}$ measure control point influence to expand geometric flexibility. Mathematically, they represent the projection of a $(d+1)$ -dimensional B-spline into a d -dimensional NURBS, with w_i derived from the original control point position $\mathbf{B}_i^w$ in the projective coordinate.

2.4.2 Isogeometric element formulation

To solve the weak formulation in Equation (3.1.8), the body $\mathcal{B}$ is discretized into elements $\mathcal{B}_E$ via the assembly operator

$$\mathcal{B} = \biguplus_{E=1}^N \mathcal{B}_E. \quad (2.4.11)$$

Elements are defined in the parametric space by adjacent knot spans. Analogous to FEA, coordinates, displacement fields, and test functions are computed from NURBS basis functions and control point displacements via

$$\mathbf{x} = \sum_{A=1}^{n_{cp}} R_A \mathbf{x}_A, \quad (2.4.12)$$

$$\mathbf{u} = \sum_{A=1}^{n_{cp}} R_A \mathbf{u}_A, \quad (2.4.13)$$

$$\boldsymbol{\eta} = \sum_{A=1}^{n_{cp}} R_A \delta \mathbf{u}_A. \quad (2.4.14)$$

However, evaluating a single element requires considering all control points whose corresponding basis functions have support over that element. The number of interacting entities is given by $n_{nes} = \prod_{i=1\dots 3}(p_i + 1)$, yielding

$$\mathbf{x}^e = \sum_{a=1}^{n_{nes}} R_a \mathbf{x}_a, \quad (2.4.15)$$

$$\mathbf{u}^e = \sum_{a=1}^{n_{nes}} R_a \mathbf{u}_a, \quad (2.4.16)$$

$$\boldsymbol{\eta}^e = \sum_{a=1}^{n_{nes}} R_a \delta \mathbf{u}_a. \quad (2.4.17)$$

Unlike FEA where n_{nes} equals the element node count, IGA's non-interpolatory nature means control points $\mathbf{x}_A$ rarely coincide with the physical shape $\mathbf{x}$, whereas $\mathbf{x}_A = \mathbf{x}$ always holds at FEA nodes. Numerical integration relies on sufficiently accurate classical GAUSS-LEGENDRE quadrature, with half-point rule alternatives detailed in [15].

2.5 Material model

For the governing equations, a compressible NEO-HOOKEAN hyperelastic model is adopted for large deformations, defining the strain energy density function Ψ as

$$\Psi = \frac{\mu}{2}(I_1 - 3 - 2 \ln J) + \frac{\lambda}{4}(J^2 - 1 - 2 \ln J), \quad (2.5.1)$$

where $I_1 = \text{tr}(\mathbf{C})$ denotes the first invariant of the right CAUCHY-GREEN tensor $\mathbf{C}$, and $J = \det(\mathbf{F})$ is the Jacobian of the deformation gradient $\mathbf{F}$.

The LAMÉ parameters μ and λ are typically parameterized via equivalent small-strain YOUNG's modulus E and POISSON's ratio ν as

$$\mu = \frac{E}{2(1 + \nu)} \quad \text{and} \quad \lambda = \frac{E\nu}{(1 + \nu)(1 - 2\nu)}, \quad (2.5.2)$$

from which the second PIOLA-KIRCHHOFF stress $\mathbf{S} = 2\partial\Psi/\partial\mathbf{C}$ is derived and pushed forward to the spatial CAUCHY stress tensor $\boldsymbol{\sigma}$.

3 Discretization Methods

In the field of mechanics, complex system behaviors are governed by Partial Differential Equations (PDEs). Because these balance laws are typically too intricate to be solved analytically, numerical approximations are required. Discretization is the core technique used in these numerical methods to enable computation. It involves breaking down a continuous problem into manageable, finite components: spatial discretization divides the physical space into discrete elements or particles, while time discretization breaks the continuous flow of time into distinct steps. Solving problems in continuum mechanics generally requires a combination of both techniques. This chapter outlines the fundamental principles of these discretization strategies. For comprehensive details, readers may consult references [4, 19, 32, 33].

3.1 Spatial Discretization

As previously noted, spatial discretization focuses on approximating solutions for PDEs without needing to calculate a global solution across an infinite number of spatial points. Instead, the overall structure is partitioned into smaller, localized domains or intervals. Within these subdomains, complex field equations can be approximated using simpler mathematical functions. Broadly speaking, spatial discretization techniques fall into three main classifications: mesh-based methods, mesh-free methods, and hybrid methods.

3.1.1 Mesh-Based Methods

Mesh-based techniques depend on generating a grid or mesh that partitions the physical region into specific discrete geometries. Common element shapes include triangles and quadrilaterals in 2D spaces, and tetrahedra or hexahedra in 3D spaces. Prominent examples of mesh-based spatial discretization include the Finite Difference Method (FDM) [17], Finite Volume Method (FVM) [10], Boundary Element Method (BEM) [8], and the FEM [4]. Because FEM is the standard and most prevalent approach for solving solid mechanics problems, it is the primary method employed in this thesis. A brief overview of FEM, starting with the governing PDEs of a continuum body, is provided below.

Finite element method

In the finite element method FEM, a partial differential equation describes the behavior of the continuum body being modeled firstly. This equation is formulated in the strong form, which strictly applies to every infinitesimal material point within the physical domain. Because solving these highly continuous differential equations analytically is often unfeasible, they are transformed into a weak form via integration techniques. The weak form is mathematically less demanding, as it reduces the required order of differentiation. To solve this weak form numerically, the global domain is partitioned into a finite number of discrete subdomains, known as finite elements. By determining the solution at the

vertices of these elements referred to as nodes the continuous field variable can be interpolated across the entire element. In the context of structural mechanics and this thesis, the primary field variable is typically the displacement field.

The strong form of the dynamic equilibrium equation is expressed as

$$\rho \ddot{\mathbf{u}} - \nabla \cdot \boldsymbol{\sigma} - \mathbf{b} = \mathbf{0}, \quad (3.1.1)$$

where $\ddot{\mathbf{u}}$ denotes the acceleration vector. To obtain a unique solution, appropriate boundary conditions must be enforced on the domain's boundary $\partial\mathcal{B}$. Every point on the boundary must be subjected to either a prescribed displacement or a prescribed traction. Prescribed displacements and tractions are mathematically defined as

$$\mathbf{u} = \hat{\mathbf{u}} \quad \text{on } \partial\mathcal{B}_D, \quad (3.1.2a)$$

$$\mathbf{t} = \boldsymbol{\sigma} \cdot \mathbf{n} \quad \text{on } \partial\mathcal{B}_N. \quad (3.1.2b)$$

To derive the weak form, Equation (3.1.1) is multiplied by an arbitrary test function (or virtual displacement) $\delta\mathbf{u}$. This test function must satisfy the homogeneous condition on the DIRICHLET boundary

$$\delta\mathbf{u} = \mathbf{0} \quad \text{on } \partial\mathcal{B}_D. \quad (3.1.3)$$

Provided that the test function remains arbitrary everywhere else, integrating the product over the entire volume $\mathcal{B}$ yields the initial weak formulation

$$\int_{\mathcal{B}} \delta\mathbf{u} \cdot \rho \ddot{\mathbf{u}} \, dv - \int_{\mathcal{B}} \delta\mathbf{u} \cdot (\nabla \cdot \boldsymbol{\sigma}) \, dv - \int_{\mathcal{B}} \delta\mathbf{u} \cdot \mathbf{b} \, dv = 0. \quad (3.1.4)$$

By applying the product rule for differentiation, the stress divergence term can be expanded as

$$\delta\mathbf{u} \cdot (\nabla \cdot \boldsymbol{\sigma}) = \nabla \cdot (\boldsymbol{\sigma} \delta\mathbf{u}) - \boldsymbol{\sigma} : \nabla \delta\mathbf{u}. \quad (3.1.5)$$

Subsequently, the application of the GAUSS divergence theorem allows the volume integral of the divergence to be converted into a surface integral

$$\int_{\mathcal{B}} \nabla \cdot (\boldsymbol{\sigma} \delta\mathbf{u}) \, dv = \int_{\partial\mathcal{B}} (\mathbf{n} \cdot \boldsymbol{\sigma}) \delta\mathbf{u} \, da. \quad (3.1.6)$$

Utilizing CAUCHY's stress theorem ($\mathbf{t} = \boldsymbol{\sigma} \mathbf{n}$) and decomposing the boundary into its DIRICHLET and NEUMANN parts, the weak form can be reformulated as

$$\int_{\mathcal{B}} \delta\mathbf{u} \cdot \rho \ddot{\mathbf{u}} \, dv + \int_{\mathcal{B}} \boldsymbol{\sigma} : \nabla \delta\mathbf{u} \, dv - \int_{\partial\mathcal{B}=\partial\mathcal{B}_D \cup \partial\mathcal{B}_N} \delta\mathbf{u} \cdot \mathbf{t} \, da - \int_{\mathcal{B}} \delta\mathbf{u} \cdot \mathbf{b} \, dv = 0. \quad (3.1.7)$$

Knowing that the virtual displacement $\delta\mathbf{u}$ is strictly zero on $\partial\mathcal{B}_D$, the surface integral over $\partial\mathcal{B}_D$ vanishes. The final weak form is thus given by

$$\int_{\mathcal{B}} \delta\mathbf{u} \cdot \rho \ddot{\mathbf{u}} \, dv + \int_{\mathcal{B}} \boldsymbol{\sigma} : \nabla \delta\mathbf{u} \, dv - \int_{\partial\mathcal{B}_N} \delta\mathbf{u} \cdot \mathbf{t} \, da - \int_{\mathcal{B}} \delta\mathbf{u} \cdot \mathbf{b} \, dv = 0. \quad (3.1.8)$$

This weak form serves as the fundamental basis for the spatial discretization in the finite element method, which generally involves four primary steps:

- **Mesh generation:** The continuous physical body is subdivided into a finite assembly of elements.

- **Spatial interpolation:** The continuous displacement field is approximated by evaluating the unknown displacements solely at the elemental nodes. Within the framework of the isoparametric concept, shape functions $N_i(\xi, \eta, \zeta)$ are introduced to interpolate the displacement field within the element's local coordinates.

$$\mathbf{u} = \mathbf{N}\mathbf{u}^e \quad \text{and} \quad \delta\mathbf{u} = \mathbf{N}\delta\mathbf{u}^e, \quad (3.1.9)$$

where $\mathbf{u}^e$ is the element e vector of nodal displacements and $\delta\mathbf{u}^e$ is the vector of virtual nodal displacements. Their gradients are

$$\nabla\mathbf{u} = \nabla\mathbf{N}\mathbf{u}^e = \mathbf{B}\mathbf{u}^e \quad \text{and} \quad \nabla\delta\mathbf{u} = \nabla\mathbf{N}\delta\mathbf{u}^e = \mathbf{B}\delta\mathbf{u}^e, \quad (3.1.10)$$

where $\mathbf{B} = \nabla\mathbf{N}$ is the spatial gradient matrix of the shape functions, commonly referred to as the strain-displacement matrix.

- **Integral evaluation and assembling.** Substituting Equation (3.1.9) and Equation (3.1.10) into Equation (3.1.8), the weak form evaluated element-wise and assembled reads

$$\delta\mathbf{u}^e \left[\int_{\mathcal{B}^e} \mathbf{N}^T \rho \mathbf{N} \ddot{\mathbf{u}}^e dv + \int_{\mathcal{B}^e} \mathbf{B}^T \boldsymbol{\sigma} dv - \int_{\partial\mathcal{B}_N^e} \mathbf{N}^T \mathbf{t} da - \int_{\mathcal{B}^e} \mathbf{N}^T \mathbf{b} dv \right] = 0, \quad (3.1.11)$$

where $\mathcal{B}^e$ denotes the physical domain of element e , and $\partial\mathcal{B}_N^e$ is the portion of the element boundary subjected to prescribed tractions.

- **Integrals evaluation and assembling.** Substituting Equation (3.1.9) and Equation (3.1.10) into Equation (3.1.8), the weak form evaluated element-wise and assembled reads

$$\delta\mathbf{u}^e \left[\int_{\mathcal{B}^e} \mathbf{N}^T \rho \mathbf{N} \ddot{\mathbf{u}}^e dv + \int_{\mathcal{B}^e} \mathbf{B}^T \boldsymbol{\sigma} dv - \int_{\partial\mathcal{B}_N^e} \mathbf{N}^T \mathbf{t} da - \int_{\mathcal{B}^e} \mathbf{N}^T \mathbf{b} dv \right] = 0. \quad (3.1.12)$$

Since Equation (3.1.12) must hold for any arbitrary $\delta\mathbf{u}^e$ satisfying Equation (3.1.3), it yields

$$\int_{\mathcal{B}^e} \mathbf{N}^T \rho \mathbf{N} \ddot{\mathbf{u}}^e dv + \int_{\mathcal{B}^e} \mathbf{B}^T \boldsymbol{\sigma} dv - \int_{\partial\mathcal{B}_N^e} \mathbf{N}^T \mathbf{t} da - \int_{\mathcal{B}^e} \mathbf{N}^T \mathbf{b} dv = 0. \quad (3.1.13)$$

For an arbitrary integrand $f(x, y, z)$ in Equation (3.1.13) expressed in local shape functions, the integration is performed by

$$\int_{\mathcal{B}^e} f(x, y, z) dv = \int_{-1}^{+1} \int_{-1}^{+1} \int_{-1}^{+1} f(\xi, \eta, \zeta) \det(\mathbf{J}(\xi, \eta, \zeta)) d\xi d\eta d\zeta, \quad (3.1.14)$$

with $\det(\mathbf{J})$ representing the Jacobian determinant, numerical integration is evaluated via

$$\begin{aligned} & \int_{-1}^{+1} \int_{-1}^{+1} \int_{-1}^{+1} f(\xi, \eta, \zeta) \det(\mathbf{J}(\xi, \eta, \zeta)) d\xi d\eta d\zeta \\ &= \sum_{i=1}^{n_s} \sum_{j=1}^{n_s} \sum_{k=1}^{n_s} f(\xi_i, \eta_j, \zeta_k) \det(\mathbf{J}(\xi_i, \eta_j, \zeta_k)) w_i w_j w_k, \end{aligned} \quad (3.1.15)$$

with integration order n_s , sampling points ξ_i, η_j, ζ_k , and weights w_i, w_j, w_k . As the most efficient scheme in FEM compared to trapezoidal or Simpson's rules, GAUSS

quadrature achieves exact integration for polynomials up to order $2n_s - 1$, with points and weights detailed in references such as [2].

Global quantities are obtained by assembling element-wise integrations

$$\mathbf{M}\ddot{\mathbf{u}} = \bigwedge_{e=1}^{n_e} \int_{\mathcal{B}^e} \mathbf{N}^T \rho \mathbf{N} \ddot{\mathbf{u}}^e dv, \quad (3.1.16a)$$

$$\mathbf{f}_{int} = \bigwedge_{e=1}^{n_e} \int_{\mathcal{B}^e} \mathbf{B}^T \boldsymbol{\sigma} dv, \quad (3.1.16b)$$

$$\mathbf{f}_{ext} = \bigwedge_{e=1}^{n_e} \left[\int_{\partial \mathcal{B}_N^e} \mathbf{N}^T \mathbf{t} da + \int_{\mathcal{B}^e} \mathbf{N}^T \mathbf{b} dv \right], \quad (3.1.16c)$$

where n_e is the total element count, $\mathbf{M}$ is the global mass matrix, $\mathbf{f}_{int}(\mathbf{u})$ represents the internal force vector, and $\mathbf{f}_{ext}$ denotes the external force vector, allowing Equation (3.1.8) to be expressed as

$$\mathbf{M}\ddot{\mathbf{u}} + \mathbf{f}_{int} - \mathbf{f}_{ext} = \mathbf{0}. \quad (3.1.17)$$

- **Solving the system.** Due to the nonlinear dependence of $\mathbf{f}_{int}$ on $\mathbf{u}$, Equation (3.1.17) is solved iteratively via the NEWTON-RAPHSON method [16]. Assuming $\mathbf{f}_{ext}$ is independent of $\mathbf{u}$, a residual vector is defined as

$$\mathbf{r}(\mathbf{u}) = \mathbf{M}\ddot{\mathbf{u}} + \mathbf{f}_{int}(\mathbf{u}) - \mathbf{f}_{ext}. \quad (3.1.18)$$

Applying a linear Taylor expansion for iteration step i yields

$$\mathbf{r}(\mathbf{u}^{(i)} + \Delta \mathbf{u}^{(i+1)}) \approx \mathbf{r}(\mathbf{u}^{(i)}) + \frac{\partial \mathbf{r}}{\partial \mathbf{u}^{(i)}} \Delta \mathbf{u}^{(i+1)} = \mathbf{0}. \quad (3.1.19)$$

The total stiffness matrix $\mathbf{K}$, comprising the mass-dependent term and the tangential stiffness $\mathbf{K}_t$, is defined as

$$\mathbf{K}(\mathbf{u}^{(i)}) = \frac{\partial \mathbf{r}}{\partial \mathbf{u}^{(i)}} = \mathbf{M} \frac{\partial \ddot{\mathbf{u}}^{(i)}}{\partial \mathbf{u}^{(i)}} + \mathbf{K}_t. \quad (3.1.20)$$

Substituting Equation (3.1.20) into Equation (3.1.19) provides the linear system

$$\mathbf{K}(\mathbf{u}^{(i)}) \Delta \mathbf{u}^{(i+1)} = \mathbf{f}_{ext} - \mathbf{M}\ddot{\mathbf{u}}^{(i)} - \mathbf{f}_{int}(\mathbf{u}^{(i)}). \quad (3.1.21)$$

The displacement is updated via

$$\mathbf{u}^{(i+1)} = \mathbf{u}^{(i)} + \Delta \mathbf{u}^{(i+1)}, \quad (3.1.22)$$

until the residual norm $\|\mathbf{r}\|$ satisfies a specified tolerance.

While NEWTON-RAPHSON offers quadratic convergence, deriving the exact tangent stiffness matrix can be highly complex or unavailable for certain sophisticated models. To avoid full linearization or to mitigate high computational costs, alternative schemes like the modified NEWTON-RAPHSON or BFGS methods are often utilized [4, 24]. Furthermore, solving the dynamic system requires a time discretization scheme to update the transient quantities over time, as discussed in section 3.2. The chosen time integration scheme subsequently provides the specific relationship for $\partial \ddot{\mathbf{u}} / \partial \mathbf{u}$, which is necessary to evaluate the effective dynamic stiffness matrix $\mathbf{K}$ during the iterative process.

3.1.2 Mesh-Free Methods

While conventional mesh-based methods like FEM struggle with severe mesh distortion and entanglement under extreme deformations, fragmentation, or penetration, pure mesh-free techniques, such as Smoothed Particle Hydrodynamics (SPH) [11] and mesh-free GALERKIN methods [5, 18], eliminate mesh tangling by using unconnected nodes. Nevertheless, they suffer from high computational costs and complex essential boundary enforcement [6, 20]. Since this work aims to mitigate mesh distortion while preserving computational efficiency, pure mesh-free methods are excluded in favor of hybrid techniques that combine mesh-based and mesh-free advantages.

3.1.3 Hybrid Methods

Hybrid methods, notably the MPM central to this thesis, integrate particles and a mesh by discretizing the material domain into moving particles that store history-dependent state variables (mass, momentum, deformation, and stress), while a fixed background grid handles displacement equations without retaining permanent data, as detailed in Chapter 4.

3.2 Time Integration and Kinematic Advection

As established in section 3.1.1, the evaluation of the effective stiffness matrix and the solution of the non-linear dynamic system require the acceleration vector to be rigorously defined. This is achieved through a specific time discretization scheme. The time discretization scheme partitions the continuous temporal domain into discrete time steps and advances the kinematic states sequentially. Depending on the stability criteria and desired computational efficiency, these integration methods can be broadly categorized as explicit or implicit.

In this work, to ensure unconditional stability and to accommodate the large time increments required for specific non-linear phenomena, an implicit NEWMARK time integration scheme is employed to drive the Eulerian background grid equations, followed by a dedicated hybrid advection scheme to update the Lagrangian material points.

3.2.1 Newmark Method

The NEWMARK- β method [22] is widely adopted for structural dynamics. At the current time step $(t + 1)$, it defines the grid displacement field as

$$\mathbf{u}_{(t+1)} = \mathbf{u}_{(t)} + \Delta t \dot{\mathbf{u}}_{(t)} + \Delta t^2 \left[\beta \ddot{\mathbf{u}}_{(t+1)} + (0.5 - \beta) \ddot{\mathbf{u}}_{(t)} \right], \quad (3.2.1)$$

and the velocity field as

$$\dot{\mathbf{u}}_{(t+1)} = \dot{\mathbf{u}}_{(t)} + \Delta t \left[\gamma \ddot{\mathbf{u}}_{(t+1)} + (1 - \gamma) \ddot{\mathbf{u}}_{(t)} \right], \quad (3.2.2)$$

where $\Delta t = t_{(t+1)} - t_{(t)}$ is the time increment, and β and γ are the algorithmic parameters. From Equation (3.2.1), the acceleration at time $(t + 1)$ can be expressed as

$$\ddot{\mathbf{u}}_{(t+1)} = \frac{1}{\beta \Delta t^2} \mathbf{u}_{(t+1)} - \tilde{\ddot{\mathbf{u}}}_{(t+1)}, \quad (3.2.3)$$

with the auxiliary historical term

$$\tilde{\mathbf{u}}_{(t+1)} = \frac{1}{\beta\Delta t^2}\mathbf{u}_{(t)} + \frac{1}{\beta\Delta t}\dot{\mathbf{u}}_{(t)} + \left(\frac{1}{2\beta} - 1\right)\ddot{\mathbf{u}}_{(t)}. \quad (3.2.4)$$

Substituting Equation (3.2.3) into Equation (3.2.2) yields the updated velocity:

$$\dot{\mathbf{u}}_{(t+1)} = \frac{\gamma}{\beta\Delta t}\mathbf{u}_{(t+1)} - \tilde{\mathbf{u}}_{(t+1)}, \quad (3.2.5)$$

with

$$\mathring{\mathbf{u}}_{(t+1)} = \frac{\gamma}{\beta\Delta t}\mathbf{u}_{(t)} - \left(1 - \frac{\gamma}{\beta}\right)\dot{\mathbf{u}}_{(t)} - \Delta t\left(1 - \frac{\gamma}{2\beta}\right)\ddot{\mathbf{u}}_{(t)}. \quad (3.2.6)$$

Consequently, the effective dynamic stiffness matrix required for the NEWTON-RAPHSON iteration is evaluated as

$$\mathbf{K}(\mathbf{u}_{(t+1)}^{(i)}) = \frac{\partial \mathbf{r}}{\partial \mathbf{u}_{(t+1)}^{(i)}} = \frac{1}{\beta\Delta t^2}\mathbf{M} + \mathbf{K}_t, \quad (3.2.7)$$

where $\mathbf{K}_t$ is the standard tangent stiffness, and the mass matrix contribution originates from the relation

$$\frac{\partial \ddot{\mathbf{u}}_{(t+1)}^{(i)}}{\partial \mathbf{u}_{(t+1)}^{(i)}} = \frac{1}{\beta\Delta t^2}\mathbf{I}. \quad (3.2.8)$$

To ensure both unconditional stability and second-order consistency without inducing algorithmic damping, the parameters are strictly chosen as $\beta = 0.25$ and $\gamma = 0.5$ [4, 13].

3.2.2 Hybrid PIC and FLIP Kinematic Advection for Material Points

Once the global dynamic equilibrium is reached and the grid nodal kinematics, namely $\mathbf{u}_{t+1}$, $\dot{\mathbf{u}}_{t+1}$ and $\ddot{\mathbf{u}}_{t+1}$, are resolved via the Newmark scheme, these state variables must be advected back to the discrete material points during the Grid to Particle (G2P) phase. In transient MPM, this mapping must be handled delicately to prevent excessive numerical dissipation or nonphysical ringing caused by particles crossing cell boundaries.

To mitigate these phenomena, a blended scheme coupling the Particle-in-Cell (PIC) and Fluid-Implicit-Particle (FLIP) mechanisms is employed. Subsequently, the particle velocity is evaluated using two distinct paradigms.

The pure PIC method evaluates the particle velocity $\dot{\mathbf{u}}_p^{PIC}$ by directly interpolating the converged grid velocity field utilizing the active generalized basis functions Φ_I , representing either standard elements or NURBS:

$$\dot{\mathbf{u}}_p^{PIC} = \sum_I \Phi_I \dot{\mathbf{u}}_{I,t+1} \quad (3.2.9)$$

While the PIC formulation strongly dampens spurious high frequency modes, its highly dissipative nature tends to artificially smear the kinetic energy over successive steps. Conversely, the pure FLIP velocity preserves the kinematic history by explicitly integrating the interpolated grid acceleration:

$$\dot{\mathbf{u}}_p^{FLIP} = \dot{\mathbf{u}}_{p,t} + \Delta t \sum_I \Phi_I \ddot{\mathbf{u}}_{I,t+1} \quad (3.2.10)$$

The FLIP method is significantly less dissipative and perfectly conserves energy; however, it is highly susceptible to cell crossing noise, known as the “ringing” effect. To optimize thermodynamic consistency and numerical stability, the ultimate particle velocity $\dot{\mathbf{u}}_{p,t+1}$ is determined via a linear convex combination:

$$\dot{\mathbf{u}}_{p,t+1} = \alpha \dot{\mathbf{u}}_p^{FLIP} + \beta \dot{\mathbf{u}}_p^{PIC} \quad (3.2.11)$$

where $\alpha \in [0, 1]$ serves as the numerical blending parameter and β equals $1 - \alpha$. The parameter α is typically constrained within $0.95 \sim 0.99$ to favor energy conservation while filtering high frequency noise. This advected velocity serves as the initial momentum carrier for the subsequent temporal increment.

4 The Material Point Method

4.1 Implicit MPM

The MPM formulation parallels the FEM. Since material points function similar to GAUSS points, the MPM conceptually represents an FEM with moving integration points [12]. These material points traverse a computational mesh and carry all state variables, including mass, volume, kinematics, stresses, external forces, and history parameters. During each time step, grid elements containing material points activate to construct a finite element problem. Initially, in the Particle to Grid (P2G) phase, mass, momentum, inertia, and external forces are mapped to the nodes. The deformation gradient and history variables remain at the material points to evaluate stresses and material tensors. After solving the system, nodal values are interpolated via shape functions to update the material point kinematics through a G2P mapping. Finally, the background mesh resets to its undeformed configuration, and this cycle repeats iteratively as depicted in Figure 4.1.1.

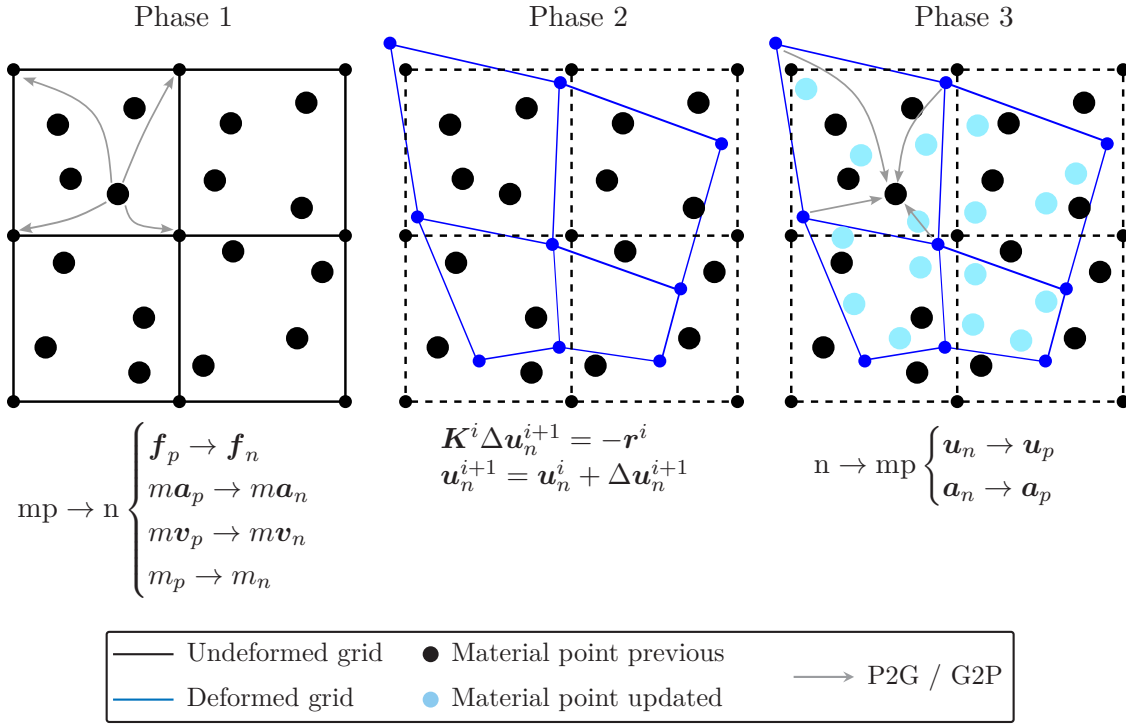


Figure 4.1.1: Phases of a computational cycle in implicit MPM.

In the first phase, state variables $\square$ are mapped from material points to grid nodes via

$$\square_n = \sum_{p=1}^{np} \mathbf{N}(\xi_p)^T \square_p, \quad (4.1.1)$$

where $\mathbf{N}$ represents the shape function matrix, ξ_p denotes the local coordinate of material point p , and np is the elemental material point count. Nodal velocity and acceleration

vectors are derived from the mapped momentum and inertia scaled by the inverse mass matrix

$$\mathbf{v}_n = \frac{(m\mathbf{v})_n}{m_n}, \quad (4.1.2a)$$

$$\mathbf{a}_n = \frac{(m\mathbf{a})_n}{m_n}. \quad (4.1.2b)$$

The second phase computes unknown nodal displacements iteratively

$$\mathbf{K}^i \Delta \mathbf{u}_n^{i+1} = -\mathbf{r}^i, \quad (4.1.3a)$$

$$\mathbf{u}_n^{i+1} = \mathbf{u}_n^i + \Delta \mathbf{u}_n^{i+1}. \quad (4.1.3b)$$

This stage strictly follows the FEM procedures detailed in section 3.1.1, substituting GAUSS points with material points for numerical integration according to Equation (3.1.15).

The third phase maps the kinematics back to the material points. To mitigate numerical dissipation and non-physical ringing, a specific projection scheme, such as the blended PIC/FLIP approach detailed in Section 3.2.2 is utilized to update the particle quantities

$$\square_p = \mathbf{N}(\boldsymbol{\xi}_p) \square_n, \quad (4.1.4)$$

and resets the computational mesh to its undeformed configuration as illustrated in Figure 4.1.1.

4.1.1 Formulation of nonlinear implicit MPM

As the MPM functions mathematically as an FEM with moving integration points, its discretized equilibrium weak form perfectly matches Equation (3.1.12). Material points simply replace standard GAUSS points for numerical integration.

Formulating the Equilibrium State

Defining the equilibrium weak form requires specifying a reference configuration. Standard finite element approaches typically employ either a Total Lagrangian (TL) formulation based on the initial configuration or an Updated Lagrangian (UL) formulation based on the current configuration. However, the TL formulation is unsuitable for the MPM because initially active grid elements may become inactive as particles move. While custom mappings could adapt the TL approach, they introduce unnecessary computational overhead compared to formulations such as UL or **PC!** (**PC!**) [9]. Consequently, only the UL framework is examined in detail here.

Prior to detailing these formulations, the deformation gradient update must be established. The deformation gradient at the start of time step t , denoted as ${}_t\mathbf{F}$, is stored as a material point history variable. The current deformation gradient $\mathbf{F}$ is computed as

$$\mathbf{F} = \Delta \mathbf{F} {}_t\mathbf{F}, \quad (4.1.5)$$

where $\Delta \mathbf{F}$ represents the incremental deformation gradient from the beginning of the time step to the current state. Analogous to Equation (2.1.4), this increment is defined as

$$\Delta \mathbf{F} = \frac{\partial \mathbf{x}}{\partial {}_t\mathbf{X}} = \frac{\partial ({}_t\mathbf{X} + \mathbf{u}_n)}{\partial {}_t\mathbf{X}} = \mathbf{I} + \frac{\partial \mathbf{u}_n}{\partial {}_t\mathbf{X}}, \quad (4.1.6)$$

where ${}_t\mathbf{X}$ signifies the undeformed coordinates of the active elements at the start of step t , and $\mathbf{u}_n$ is the nodal displacement vector accumulating from the start of the time step to the current configuration.

UL formulation

In the UL formulation, the discretized equilibrium weak form is

$$\int_v \mathbf{N}^T \rho \mathbf{N} \ddot{\mathbf{u}}_n \, dv + \int_v \mathbf{B}^T \boldsymbol{\sigma} \, dv - \int_{\partial a} \mathbf{N}^T \mathbf{t} \, da - \int_v \mathbf{N}^T \mathbf{b} \, dv = 0, \quad (4.1.7)$$

where all quantities, including the volume v and surface ∂a , correspond to the current configuration. Incorporating geometric nonlinearity and the compressible NEO-HOOKEAN material model from section 2.5 renders Equation (4.1.7) nonlinear with respect to nodal displacements. By employing the implicit NEWMARK time discretization scheme and the NEWTON-RAPHSON method for the nonlinear resolution, the internal force vector and material point stiffness matrix are given by

$$\mathbf{f}_p = \mathbf{B}_L^T \boldsymbol{\sigma} v_p, \quad (4.1.8a)$$

$$\mathbf{K}_p = \frac{1}{\beta \Delta t^2} \mathbf{M} + \mathbf{K}_T. \quad (4.1.8b)$$

The tangential stiffness matrix $\mathbf{K}_T$ is

$$\mathbf{K}_T = \left[\mathbf{B}_L^T \mathbf{C}_T \mathbf{B}_L + \mathbf{B}_{NL}^T \tilde{\boldsymbol{\sigma}} \mathbf{B}_{NL} \right] v_p, \quad (4.1.9)$$

where v_p denotes the material point volume. For a general three-dimensional case, the global shape function derivative matrices are assembled by nodal sub-matrices as $\mathbf{B}_L = [\mathbf{B}_{L,1}, \dots, \mathbf{B}_{L,n}]$ and $\mathbf{B}_{NL} = [\mathbf{B}_{NL,1}, \dots, \mathbf{B}_{NL,n}]$. For a generic node I where $I = 1 \dots n$, the corresponding linear strain-displacement sub-matrix $\mathbf{B}_{L,I}$ is defined as

$$\mathbf{B}_{L,I} = \begin{bmatrix} N_{I,x} & 0 & 0 \\ 0 & N_{I,y} & 0 \\ 0 & 0 & N_{I,z} \\ 0 & N_{I,z} & N_{I,y} \\ N_{I,z} & 0 & N_{I,x} \\ N_{I,y} & N_{I,x} & 0 \end{bmatrix}. \quad (4.1.10)$$

By introducing the spatial gradient vector of the shape function ∇N_I , the non-linear sub-matrix $\mathbf{B}_{NL,I}$ is compactly expressed as

$$\mathbf{B}_{NL,I} = \begin{bmatrix} \nabla \mathbf{N}_I & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \nabla \mathbf{N}_I & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \nabla \mathbf{N}_I \end{bmatrix}, \quad \text{where} \quad \nabla \mathbf{N}_I = \begin{bmatrix} N_{I,x} \\ N_{I,y} \\ N_{I,z} \end{bmatrix}, \quad (4.1.11)$$

where $N_{I,x}$, $N_{I,y}$, and $N_{I,z}$ are the shape function derivatives with respect to the current spatial coordinates $\mathbf{x} = \{x, y, z\}$. These spatial gradients are updated dynamically via the chain rule, yielding

$$\frac{\partial \mathbf{N}}{\partial \mathbf{x}} = \frac{\partial \mathbf{N}}{\partial {}_t\mathbf{X}} \frac{\partial {}_t\mathbf{X}}{\partial \mathbf{x}} = \frac{\partial \mathbf{N}}{\partial {}_t\mathbf{X}} \Delta \mathbf{F}^{-1}, \quad (4.1.12)$$

where $\frac{\partial \mathbf{N}}{\partial {}_t\mathbf{X}}$ denotes the shape function derivatives with respect to the coordinates at the previous time step ${}_t\mathbf{X} = \{{}_tX, {}_tY, {}_tZ\}$, and $\Delta \mathbf{F}$ is the incremental deformation gradient.

The CAUCHY stress stems from pushing forward the second PIOLA-KIRCHHOFF stress tensor

$$\boldsymbol{\sigma} = \frac{1}{\det(\mathbf{F})} [\mathbf{F} \mathbf{S} \mathbf{F}^T] , \quad (4.1.13)$$

where the second PIOLA-KIRCHHOFF stress follows from differentiating the strain energy density function Ψ of the NEO-HOOKEAN model with respect to the right CAUCHY-GREEN deformation tensor $\mathbf{C}$

$$\mathbf{S} = 2 \frac{\partial \Psi}{\partial \mathbf{C}} , \quad (4.1.14)$$

where $\mathbf{S}$ corresponds to the respective stress measure required. Furthermore, Equation (4.1.8a) expresses the CAUCHY stress in VOIGT notation

$$\boldsymbol{\sigma} = \begin{Bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{zz} \\ \sigma_{xy} \\ \sigma_{yz} \\ \sigma_{zx} \end{Bmatrix} , \quad (4.1.15)$$

whereas Equation (4.1.9) adopts the matrix form

$$\tilde{\boldsymbol{\sigma}} = \begin{bmatrix} \bar{\boldsymbol{\sigma}} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \bar{\boldsymbol{\sigma}} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \bar{\boldsymbol{\sigma}} \end{bmatrix} , \quad \text{with} \quad \bar{\boldsymbol{\sigma}} = \begin{bmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{bmatrix} . \quad (4.1.16)$$

$\mathbf{C}_T$ denotes the tangential material stiffness tensor pushed forward to the current configuration [4].

In summary, the UL formulation is highly effective for implementation within the MPM framework. Because the NEO-HOOKEAN material model is formulated in the current configuration, employing the UL formulation proves computationally efficient, as it avoids the necessity of mapping stresses and material tensors back to a previously converged state. The construction of the non-linear strain-displacement matrices necessitates specific kinematic transformations. As established, the spatial derivatives of the shape functions with respect to the current coordinates are obtained by multiplying the material derivatives by the inverse of the deformation gradient, as shown in Equation (4.1.12).

Combining the discussions in section 3.1.1, Section 3.2.1, and section 4.1, the steps of the implicit MPM are presented in Algorithm 1.

For the formulation shown above, the local coordinates of the material points in the background mesh elements are required. Those coordinates are changing each time step and evaluated using the NEWTON-RAPHSON method. Given material point global coordinates, $\mathbf{x}_p$, and element coordinates, ${}_t\mathbf{X}$, material point local coordinates are calculated as shown in Algorithm 2.

4.2 Cell crossing error

The standard MPM suffers from cell crossing error, where material points crossing element boundaries generate incorrect stress values. As illustrated in Figure 4.2.1, elongating a bar

Algorithm 1 MPM

```
1: map data from material points to nodes, Equation (4.1.1);
2: while  $t \leq t_{max}$  do
3:   activate background grid elements;
4:   while  $\|\mathbf{r}\| > tol$  do
5:     calculate  $\ddot{\mathbf{u}}$  and  $\dot{\mathbf{u}}$ , Equation (3.2.3) and Equation (3.2.5);
6:     initiate global quantities  $\mathbf{K} = \mathbf{0}$ ,  $\mathbf{f}_{int} = \mathbf{0}$ ;
7:     for  $e = 1 \rightarrow n_{elem}$  do
8:       initiate element quantities  $\mathbf{K}_e = \mathbf{0}$ ,  $\mathbf{f}_e = \mathbf{0}$ ;
9:       for  $p = 1 \rightarrow lint$  do
10:        determine MP local coordinates and shape functions;
11:        calculate  $\Delta \mathbf{F}$  and  $\mathbf{F}$ , Equation (4.1.6) and Equation (4.1.5);
12:        evaluate  $\boldsymbol{\sigma}$  and material tangent  $\mathbf{C}_T$ , Equation (4.1.13)
        and Equation (4.1.9);
13:        evaluate  $\mathbf{B}_L$  and  $\mathbf{B}_{NL}$ , Equation (4.1.10) and Equation (4.1.11);
14:        calculate  $\mathbf{K}_p$  and  $\mathbf{f}_p$  weighted by MP volume, Equation (4.1.8);
15:        accumulate  $\mathbf{K}_p$  and  $\mathbf{f}_p$  into element arrays  $\mathbf{K}_e$  and  $\mathbf{f}_e$ ;
16:      end for
17:      assemble element arrays  $\mathbf{K}_e$  and  $\mathbf{f}_e$  into global  $\mathbf{K}$  and  $\mathbf{f}_{int}$ ;
18:    end for
19:    calculate  $-\mathbf{r} = \mathbf{f}_{ext} - \mathbf{M}\ddot{\mathbf{u}} - \mathbf{f}_{int}$ ;
20:    solve  $\mathbf{K}\Delta \mathbf{u} = \mathbf{r}$  and update nodal displacement Equation (4.1.3);
21:  end while
22:  map data from nodes to material points, Equation (4.1.4);
23:  map data from material points to nodes, Equation (4.1.1);
24: end while
```

Algorithm 2 Evaluate material point local coordinates

```
1: initiate local coordinates,  $\boldsymbol{\xi}_p = \mathbf{0}$ ;
2: initiate global coordinates,  $\bar{\mathbf{x}}_p = \mathbf{N}(\boldsymbol{\xi}_p)_t \mathbf{X}$ ;
3: calculate residual vector,  $\mathbf{r} = \mathbf{x}_p - \bar{\mathbf{x}}_{pi}$ ;
4: while  $\|\mathbf{r}\| > tol$  do
5:   update local coordinates,  $\boldsymbol{\xi}_p = \boldsymbol{\xi}_p + \left[ \frac{d\bar{\mathbf{x}}_p}{d\boldsymbol{\xi}_p} \right]^{-1} \mathbf{r}$ ;
6:   update global coordinates,  $\bar{\mathbf{x}}_p = \mathbf{N}(\boldsymbol{\xi}_p)_t \mathbf{X}$ ;
7:   calculate residual vector,  $\mathbf{r} = \mathbf{x}_p - \bar{\mathbf{x}}_{pi}$ ;
8: end while
```

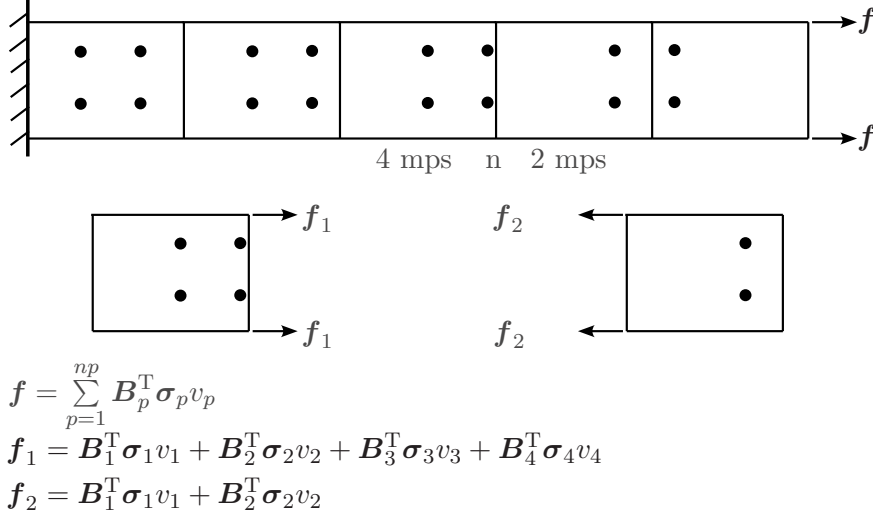


Figure 4.2.1: Material points cross elements boundaries resulting in incorrect stress values with internal forces evaluated at the global node I .

under a uniaxial tip force causes material points to cross boundaries. Maintaining equilibrium at the global node I requires equal internal forces from adjacent elements. Consequently, an unequal distribution of material points across elements forces non-physical stress variations to satisfy this equilibrium, triggering spurious stress oscillations.

Prominent mitigation strategies include particle domain interpolation techniques like the Generalized Interpolation Material Point (GIMP) and Convected Particle Domain Interpolation (CPDI) methods alongside their second-order extensions [3, 26, 23, 27]. These approaches smear the influence of a particle across adjacent elements by substituting the standard DIRAC delta function with a characteristic particle domain.

Although these methods significantly alleviate the cell-crossing error, they fundamentally rely on a background grid discretized with standard C^0 -continuous LAGRANGIAN shape functions. Consequently, spatial derivatives remain discontinuous across element boundaries. Tracking and updating geometric particle domains under extreme deformations also introduces severe numerical inaccuracies.

Enhancing the spatial continuity of the background grid offers a fundamental solution to eliminate the cell-crossing error. Replacing standard C^0 finite element shape functions with high-order smooth basis functions like B-splines or NURBS intrinsically prevents artificial stiffness and stress jumps at boundaries. Therefore, rather than employing complex particle domain updates, this thesis adopts IGA-MPM to handle large deformations and efficiently resolve the cell crossing error. Building upon the geometric fundamentals of IGA detailed in section 2.4, the subsequent section integrates B-spline basis functions directly into the MPM continuum.

4.3 IGA-MPM introduction

To practically implement this higher-order continuity and eliminate the aforementioned cell-crossing instability, MOUTSANIDIS et al. proposed the IGA-MPM [21]. This method bridges the geometric accuracy of IGA with the particle-based MPM framework. As established in section 2.4, substituting standard linear shape functions with higher-order smooth B-splines or NURBS enables IGA-MPM to construct a background approximation

space featuring intrinsic C^{p-1} continuity.

The IGA-MPM implementation follows the conventional framework but introduces essential state mapping distinctions as illustrated in Figure 4.3.1. The standard formulation shown in Figure 4.3.1a employs physical nodes and C^0 -continuous linear basis functions. This strictly localizes the projection of state variables, such as mass m_p and velocity $\mathbf{v}_p$, to the four adjacent global nodes of the host element. Conversely, the isogeometric approach depicted in Figure 4.3.1b replaces physical nodes with a generalized control point network and applies higher-order NURBS basis functions $R_A(\boldsymbol{\xi})$ formulated via the COX-DE BOOR recursion detailed in section 2.4. These functions provide a wider support stencil to expand the state mapping non-locally. Specifically, a quadratic formulation with polynomial degrees $p = q = 2$ maps a single material point's state to nine control points within the active knot span defined in section 2.4.

As conceptually compared in Figure 4.3.2, this expanded support domain provides a fundamental mathematical solution to the cell-crossing error. In standard formulations, C^0 -continuous shape functions produce abrupt spatial gradient changes at element boundaries, yielding discontinuous strain rate evaluations when particles transition between cells. By employing higher-order NURBS basis functions, IGA-MPM guarantees C^{p-1} continuity of the spatial gradients across all internal knot lines. As a material point traverses a parametric boundary, its state mapping transitions smoothly among the non-local control points rather than experiencing an abrupt reassignment. This higher-order smoothness ensures a strictly continuous representation of the deformation gradient and strain rate throughout the entire domain. Consequently, the derived stresses remain continuous, intrinsically eliminating the non-physical stress jumps and spurious oscillations without requiring complex particle domain tracking techniques.

Beyond resolving the cell-crossing instability, IGA-MPM inherits the geometric exactness of NURBS introduced in Extension to NURBS. This characteristic enables the precise representation of complex curved boundaries and all conic sections. Consequently, it eliminates the artificial boundary roughness and spurious stress concentrations typically introduced by standard piecewise-linear elements, ensuring highly accurate stress evaluations along complex curved material interfaces. The detailed procedures for integrating and generating this exact geometry within the computational framework are further discussed in Section 5.2.2.

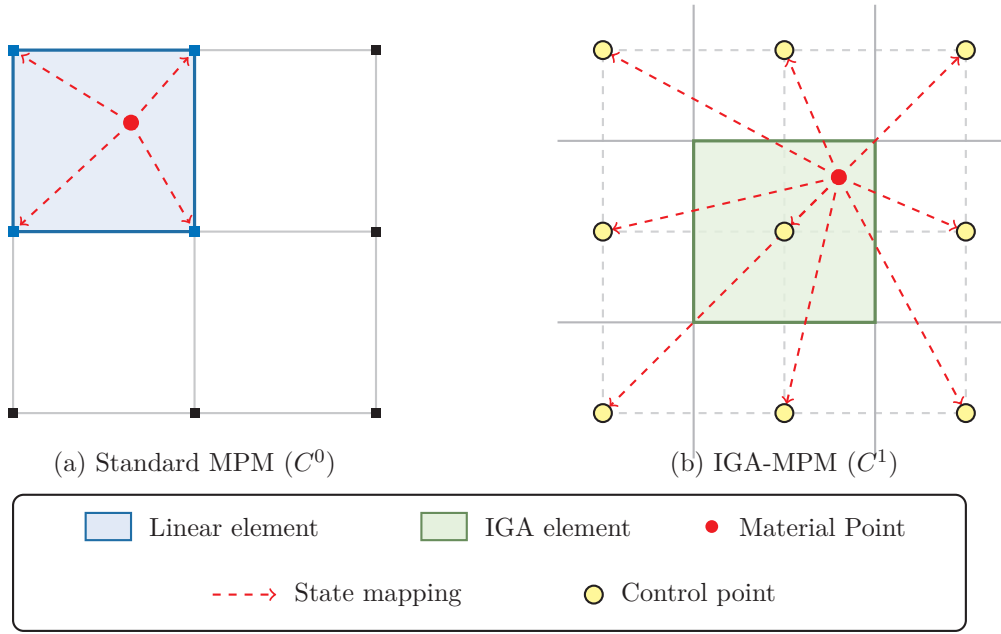


Figure 4.3.1: Comparison of state mapping between standard MPM and IGA-MPM.

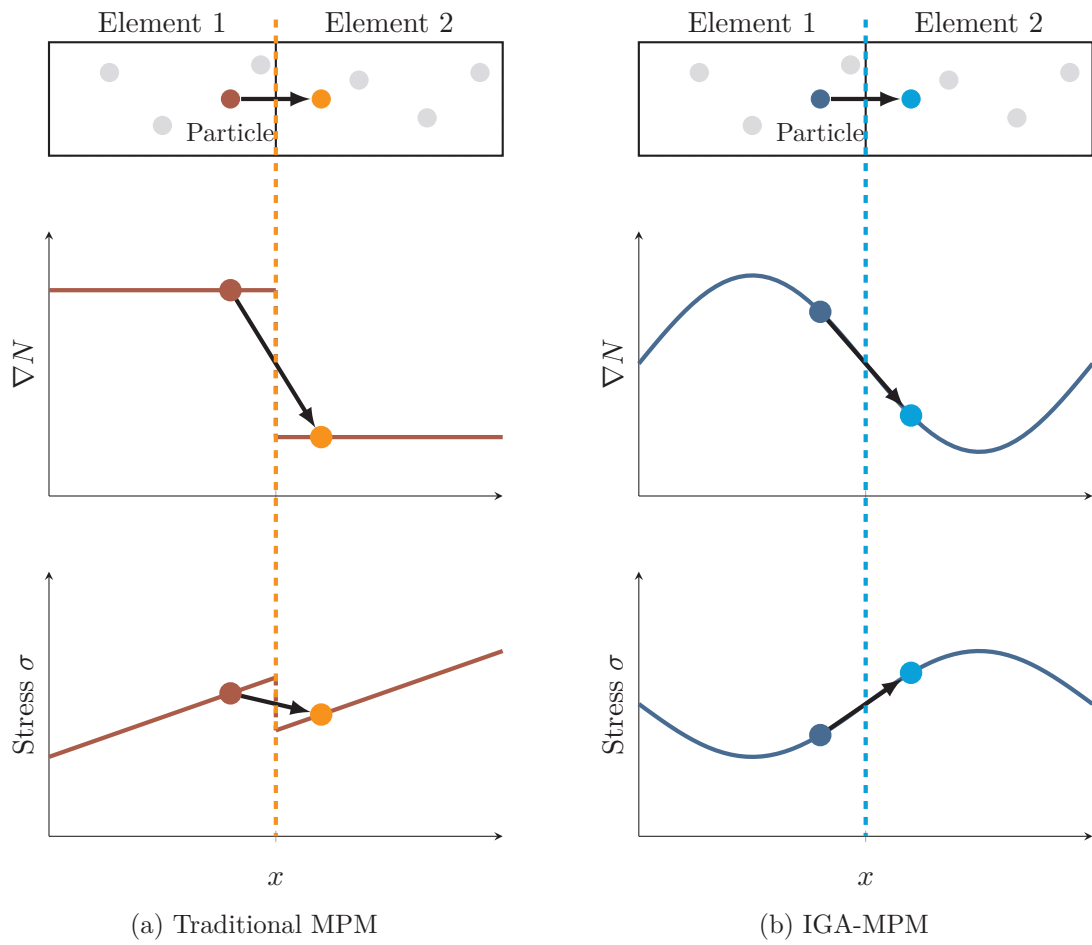


Figure 4.3.2: Comparison of the cell crossing error.

5 Algorithmic Design and Implementation within the FEAP Framework

This chapter bridges the theoretical foundations established previously with their practical computational realization. The primary objective is to implement a framework for both the standard MPM and the IGA-MPM. The chapter is structured along the logical computational workflow: material point initialization, static solution, and dynamic solution[31].

5.1 Architecture of the FEAP Framework

The FEAP, developed at the University of California, Berkeley, serves as the core computational environment for this report. Its highly extensible and modular architecture provides an ideal environment for the implementation and validation of novel numerical methods without the need to construct a standalone solver from scratch. As shown in Figure 5.1.1, the MPM framework seamlessly integrates into the native FEAP architecture, where solid blocks represent standard FEM modules and dashed blocks highlight the custom MPM additions. By embedding particle initialization alongside P2G and G2P mappings within the standard time step loop, a fully coupled computational framework is achieved. Furthermore, building upon this foundational architecture and following the mathematical definitions provided in the isogeometric user documentation [30], the framework incorporates a specialized IGA module. Instead of relying on intermediate transformations such as Bézier extraction, this implementation operates directly on the fundamental B-spline data structures by accessing internal patch data and knot vector arrays to evaluate shape functions via the Cox-de Boor recursion. Because these high-order basis functions provide C^{p-1} continuity across knot spans, this direct evaluation intrinsically smooths spatial gradients during particle transitions, thereby fundamentally eliminating the cell-crossing errors inherent to standard MPM.

5.2 Material Point Initialization

The initial configuration of the computational domain is established differently depending on whether standard discrete elements or isogeometric B-spline structures are employed.

5.2.1 Standard MPM Initialization

Before the computational loop begins, the initial configuration is established by the `MPinit` subroutine, as schematically summarized in Figure 5.2.1. The algorithm traverses the entire mesh to distinguish between generator elements, designated as material 2, and computational background elements, designated as material 1. For each generator element, as

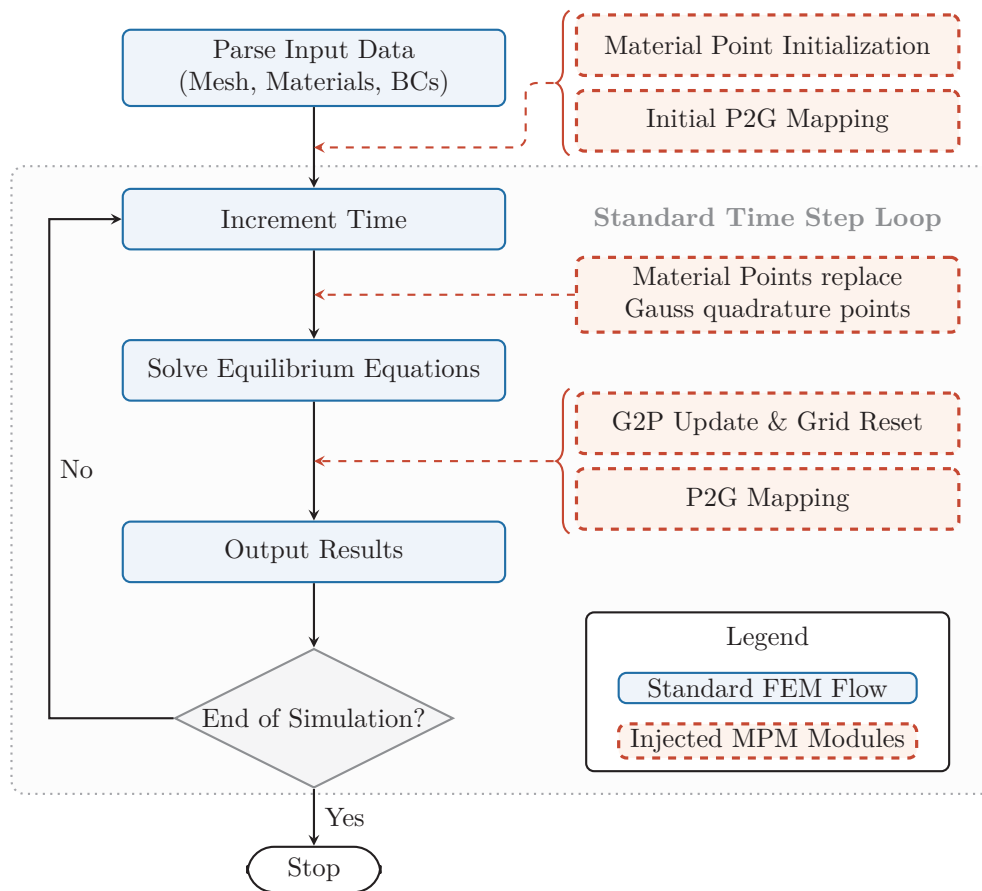


Figure 5.1.1: Integration of the MPM framework into the native FEAP architecture.

illustrated in Figure 5.2.1b, the routine uniformly distributes material points based on a user-defined one-dimensional density parameter. Using standard C^0 isoparametric shape functions, these local parametric points are mapped to global physical coordinates. Each particle's initial volume is calculated using the weighted Jacobian determinant, concurrently initializing its mass, zeroed velocity, and identity deformation gradient. Crucially, once the material points are generated, the generator elements act merely as temporary geometric templates and are subsequently deactivated, as depicted in Figure 5.2.1c. Meanwhile, the background elements remain inactive until the computational loop begins.

5.2.2 IGA-MPM Initialization

Building upon the core logic of `MPinit`, the IGA-MPM framework employs the `IGAMPinit` subroutine, which shares identical state allocation and template deactivation procedures but fundamentally differs in its geometric treatment. Instead of discrete elements and nodal arrays, `IGAMPinit` parses global knot vectors and operates on active knot spans utilizing local control points. Material points are mapped to physical coordinates using smooth B-spline basis functions rather than standard shape functions. As compared in Figure 5.2.2, this isogeometric mapping ensures that the seeded particles perfectly capture true curved boundaries without geometric discretization gaps. Finally, due to the complexity of IGA input files in FEAP, `IGAMPinit` exports the generated particle data to an external file for the main solver to read, unlike the standard MPM which integrates particle generation and subsequent calculation within a single input file.

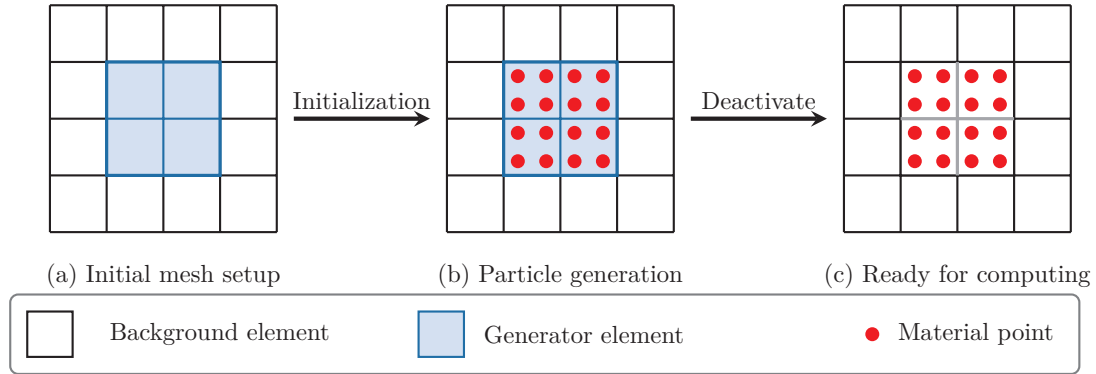


Figure 5.2.1: Schematic of the procedural material point initialization algorithm.

5.3 Algorithmic Implementation of the Computational Loop

While the mathematical and kinematic foundations of the standard MPM and IGA-MPM were established in Chapter 4, this section strictly details their algorithmic realization and integration within the FEAP computational framework. The focus is directed toward spatial searching algorithms, memory management, and data structure manipulation during the state mapping phases.

5.3.1 Algorithmic Integration and Quadrature Substitution

The standard execution logic of the FEAP is driven by a macro-command architecture. Our MPM implementation leverages this modularity to seamlessly extend existing func-

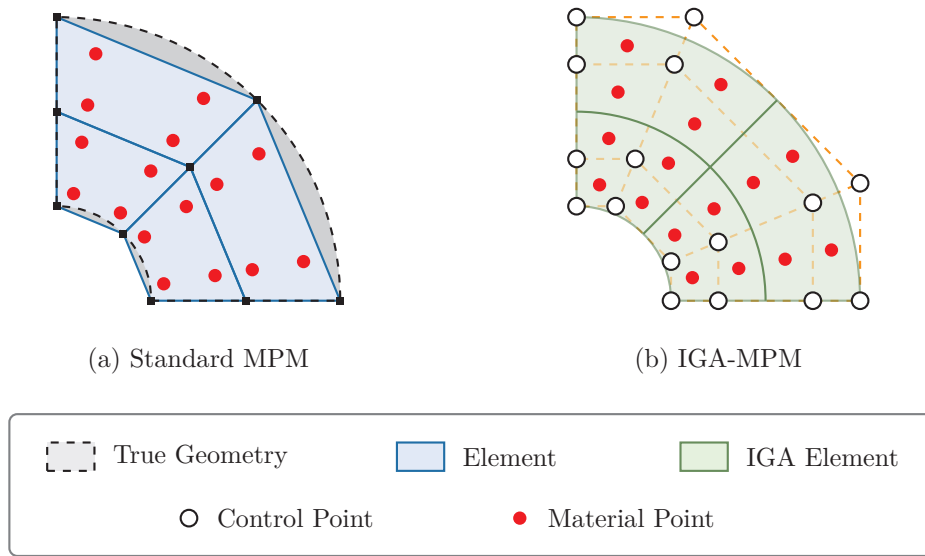


Figure 5.2.2: Comparison of material point initialization for curved geometry.

tionalities.

The integration utilizes two main modifications. First, we use custom commands to handle MPM-specific tasks independently of the standard FEM routine. These tasks include generating initial particles and mapping data between the particles and the grid. Second, we intercept the core calculation process during the global assembly phase by modifying standard quadrature subroutines, such as `quadr2d` and `quadr3d`. When the MPM solver flag is active, the conventional stationary Gauss quadrature point generation is bypassed. Instead, the solver calls `MPM_getPoints` to retrieve the moving Lagrangian material points within the current element. The particles' local parametric coordinates and initial volumes are then directly substituted as the integration points and weights, respectively. This allows the equations to be evaluated exactly where the particles are located. Furthermore, this approach extends naturally to dynamic analysis. By also evaluating the mass matrix and inertial forces at the particle locations, the framework's existing time integration algorithms can be directly utilized for dynamic simulations.

5.3.2 P2G for Standard MPM

The computational cycle initiates with the P2G phase, executed by the `P2G` subroutine. A primary prerequisite for this procedure is identifying the corresponding host element for each material point, thereby establishing the grid-particle topology required for subsequent state mapping. To achieve this efficiently, the algorithm first employs a bounding box filter to quickly identify potential host elements. Subsequently, a local Newton-Raphson search accurately determines the particle's parametric coordinates.

Upon successful topological localization, the algorithm executes a state projection utilizing standard C^0 finite element shape functions. The particle's mass, momentum, and acceleration states are systematically accumulated into dynamically allocated global background arrays. Within the specific numerical architecture of FEAP, if the implicit dynamic solver is activated, the global generalized rate arrays are explicitly overwritten with the computed nodal velocities and accelerations to drive the time integration

5.3.3 P2G for IGA-MPM

The isogeometric P2G implementation, managed by the `IGAP2G` subroutine, shares identical post-projection procedures with the standard method, specifically the accumulation of mass and inertial states, the computation of generalized nodal rates, and the enforcement of zero-velocity constraints on empty nodes. However, it fundamentally differs in topological searching and geometric mapping due to the non-local nature of B-spline patches.

Instead of utilizing bounding box filters and local Newton-Raphson iterations across discrete elements, the search algorithm operates directly on global knot vectors extracted from the FEAP topology blocks. For each particle, physical coordinates are linearly mapped to the global parametric domain, followed by a binary search to accurately identify active knot spans and normalize them into the standard parent domain. The state projection then employs high-order B-spline basis functions rather than standard finite element shape function.

5.4 Algorithmic Support for Transient MPM Simulations in FEAP

Extending the MPM framework within FEAP to support transient dynamic simulations requires modifications to the element-level matrix assembly and particle state preservation. This section details how the standard routines are adapted to compute particle-based inertial forces and how grid-based kinematic variables are mapped back to update the material point histories.

5.4.1 Unified Dynamic Mass Assembly and Inertial Forces

The transient formulation resides in modifying the element-level residual routines `f1d2d1` and `iner2d`. When the MPM solver flag is active, the conventional differential volume computation based on Gauss points is overridden. Within `f1d2d1`, the quadrature weight is directly replaced by the invariant initial particle volume. Similarly, `iner2d` substitutes the differential mass with the localized particle mass. Consequently, the mass matrix and inertial forces are assembled by looping strictly over active material points using shape functions evaluated at particle locations. This particle-centric assembly rigorously guarantees mass conservation.

5.4.2 Implementation of Kinematic Advection and State Preservation

During element state evaluations, grid-based kinematic quantities must be extracted and mapped back to the discrete particle data structures. This advection process is managed via the `MPM_kin` subroutine. This subroutine computes the incremental deformation gradient to multiplicatively advance the deformation gradient according to Equation (4.1.5), and stores the interpolated grid displacements and accelerations directly into the particles. Subsequently, following the material constitutive evaluations within the native `modlfd` routine, the updated CAUCHY stresses are extracted and saved back to each material point via the `MPM_updateStress` subroutine. This integrated data pipeline ensures that macroscopic deformation histories and internal stress variables are strictly preserved across consecutive time increments.

5.5 G2P Kinematic Update and Grid Reset

A key advantage of integrating both methods within a singular framework is the algorithmic unification of the G2P update phase executed by the `G2P` subroutine. Regardless of whether the governing equations were assembled using standard nodes or isogeometric control points, the underlying solver returns a unified multidimensional generalized displacement array. The particles fetch the corresponding increments from this generalized array to explicitly update their physical positions and preserve their historical deformation gradients. When the dynamic solver is active, particle velocities are additionally advanced utilizing a pure FLIP integration scheme based on the nodal accelerations. Once the mapping is completed, the background mesh must be reinitialized to prepare for the next iteration. Specifically, all global kinematic arrays associated with the background grid, including the total and incremental displacements along with the dynamic velocity and acceleration arrays, are explicitly reset to zero. This comprehensive reset ensures that the computational grid remains a pristine and undeformed reference frame for the subsequent time step.

6 Numerical Examples and Validation

6.1 Convergence analysis of a static cantilever bar

To evaluate the spatial and quadrature convergence of the different numerical methods, a two-dimensional static cantilever bar subjected to a body force is adopted as the benchmark test model. Following the basic setup in similar benchmarks, the bar has a length of $L = 8$ m, a height of $H = 1$ m, and an out-of-plane thickness of $t = 1$ m. The material is modeled using a standard linear elastic constitutive law with a YOUNG's modulus of $E = 10$ Pa and a POISSON's ratio of $\nu = 0.3$. A uniform downward body force of $b = 8 \times 10^{-3}$ N/m³ is incrementally applied to the domain in 10 equal load steps. To constrain the body, the left boundary of the model is fully clamped, while the remaining boundaries are free. Due to the lack of an exact analytical solution for this specific configuration, a high-resolution FEM solution is computed and utilized as reference solution for the subsequent error evaluations.

For the spatial convergence study, the background grid is uniformly refined with characteristic element sizes $h \in \{1, 0.5, 0.25, 0.125, 0.0625\}$ m. Three types of discretizations are considered: pure FEM, traditional MPM, and the proposed IGA-MPM. Both FEM and MPM utilize standard conforming C^0 -continuous bilinear shape functions (Q4), whereas the IGA-MPM employs C^1 -continuous quadratic B-splines to provide higher smoothness and mitigate cell-crossing errors. To quantify the computational accuracy, the spatial convergence of the displacement error is evaluated on the discrete material points for MPM and IGA-MPM, and on the GAUSS points for FEM. By mapping the physical coordinates of these points to the local natural coordinates (ξ, η) of the corresponding high-resolution FEM reference elements, the reference displacement $\mathbf{u}^{\text{ref}}$ is computed via shape function interpolation. From these, the relative L_2 and L_∞ error norms are computed as

$$L_2 = \frac{\sqrt{\frac{1}{N_p} \sum_{p=1}^{N_p} \|\mathbf{u}_p^{\text{num}} - \mathbf{u}_p^{\text{ref}}\|_2^2}}{\sqrt{\frac{1}{N_p} \sum_{p=1}^{N_p} \|\mathbf{u}_p^{\text{ref}}\|_2^2}}, \quad (6.1.1)$$

and

$$L_\infty = \frac{\max_{1 \leq p \leq N_p} \|\mathbf{u}_p^{\text{num}} - \mathbf{u}_p^{\text{ref}}\|_2}{\max_{1 \leq p \leq N_p} \|\mathbf{u}_p^{\text{ref}}\|_2}, \quad (6.1.2)$$

given the numerical values for the displacements $\mathbf{u}^{\text{num}}$, the interpolated reference solution $\mathbf{u}^{\text{ref}}$, and the total number of evaluation points N_p .

The spatial convergence results are shown in Figure 6.1.1a, where the number of particles per cell (Particles Per Cell (PPC)) is kept constant. To facilitate visual comparison, the theoretical first-order ($\mathcal{O}(h^1)$) and second-order ($\mathcal{O}(h^2)$) convergence slopes are provided as references. As a baseline, FEM exhibits an ideal second-order convergence rate, with the relative L_2 error decreasing steadily from 31.34% to 0.18%. In contrast, the traditional MPM shows a distinctly different convergence behavior: although it achieves second-order convergence similar to FEM at coarser meshes, it exhibits a significant lack of convergence and error plateaus upon further mesh refinement. This observation aligns well with the

cell-crossing instability phenomena documented in the literature [29]. Specifically, the L_2 error of the MPM drops to 0.81% at $h = 0.25$ m but anomalously increases to 1.14% at $h = 0.0625$ m. This non-linear, anomalous convergence behavior can mostly be attributed to the cell-crossing errors caused by the C^0 -continuous shape functions, where the discontinuous gradients at cell boundaries disrupt the integration accuracy. Conversely, IGA-MPM demonstrates a remarkable improvement, showing excellent convergence without any error plateaus. The enhanced regularity of the C^1 quadratic B-splines effectively eliminates the step-like stress oscillations and cell-crossing issues, reducing the L_2 error smoothly from 5.00% to 0.14% upon mesh refinement. Overall, IGA-MPM yields significantly smaller errors and achieves a stable convergence rate between first and second order, which is highly consistent with the findings reported by [21].

For the quadrature convergence study, the grid resolution is fixed at $h = 0.25$ m, and the number of material points per cell is gradually increased. The quadrature convergence results are depicted in Figure 6.1.1b. At the lowest particle density (PPC=4), both methods exhibit significant integration inaccuracies: the traditional MPM yields an L_2 error of 10.40%, while the IGA-MPM gives a larger error of 14.07%, due to its broader integration domain and higher polynomial order of the quadratic B-splines, which demand finer quadrature sampling. As the particle density increases, the accuracy of both methods improves substantially. For PPC=9 and above, the IGA-MPM consistently outperforms the traditional MPM; for instance, at PPC=9, the L_2 error of IGA-MPM is 1.16% compared to 2.05% for MPM. With further refinement to PPC=16 and 25, the IGA-MPM errors drop to 0.71% and 0.30%, respectively, confirming the beneficial effect of increased particle density. Overall, both methods exhibit a high convergence rate greater than second order in terms of quadrature convergence, fully demonstrating that increasing the particle density can effectively overcome integration inaccuracies and substantially improve the overall computational accuracy.

In addition to the spatial and quadrature convergence, the algorithmic convergence behavior of the nonlinear solution scheme is investigated. Figure 6.1.2 illustrates the evolution of the residual norm during the Newton Raphson iterations at load step 10. To appropriately assess the convergence rate, theoretical reference lines for linear ($\mathcal{O}(e_k)$) and quadratic ($\mathcal{O}(e_k^2)$) error decay are additionally plotted. As observed, all three methods exhibit a rapid decay, converging to a residual norm below 10^{-20} within four iterations. The convergence profiles in the semilogarithmic plot follow the theoretical quadratic convergence rate, bending downwards and clearly outperforming the linear reference boundary. While a minor deviation from the pure FEM solution is observed for both MPM and IGA-MPM at the final iterations, primarily stemming from the inherent quadrature errors introduced by particle integration. This excellent asymptotic quadratic convergence not only highlights the numerical robustness of the proposed framework under severe deformation but also verifies the exactness of the derived and implemented tangent stiffness matrices.

6.2 Pulse impact on an elastic bar

To validate the accuracy of the proposed method and to highlight the limitations of traditional approaches, we investigate the dynamic response of a one-dimensional elastic bar. A bar of length $L = 1$ m is clamped at one end, while a longitudinal transient body force is applied to the entire domain. The material is modeled using a NEO-HOOKEAN hyperelastic constitutive model with the following parameters: YOUNG's modulus $E = 1.0 \times 10^2$ Pa, POISSON's ratio $\nu = 0.0$, and density $\rho = 1.0 \times 10^{-6}$ kg/m³. The applied horizontal body

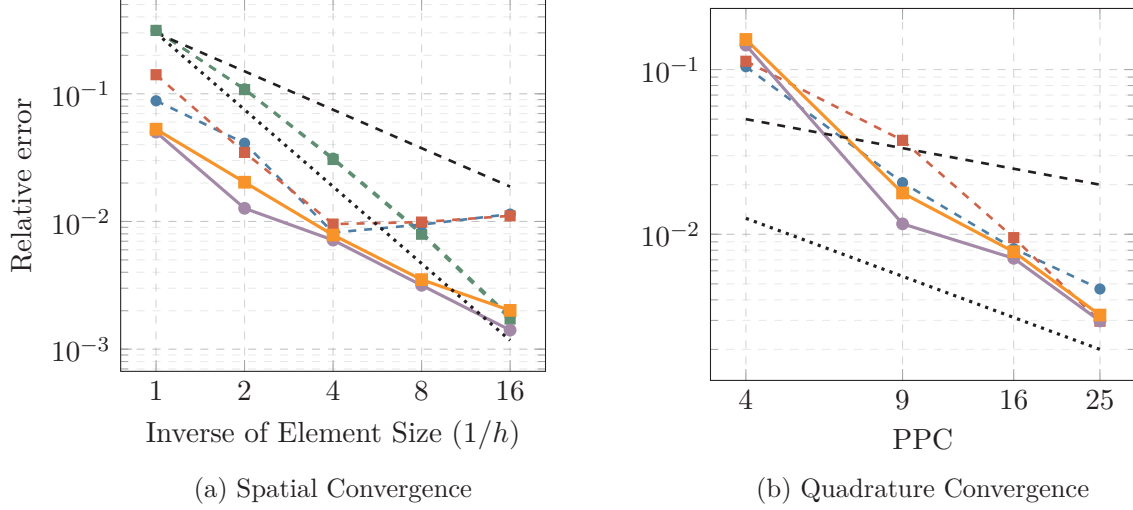


Figure 6.1.1: Convergence analysis of the static bar model.

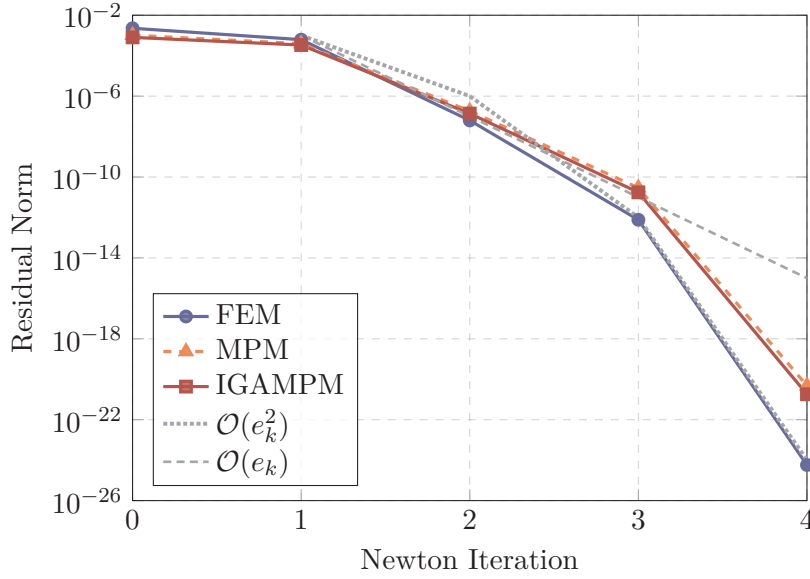


Figure 6.1.2: Comparing residual norm of FEM, traditional MPM, IGA-MPM.

force has a base magnitude of 1.0 N/m^3 , and its amplitude follows a trapezoidal pulse profile in time: it increases linearly to the maximum value at $t = 5.0 \times 10^{-5} \text{ s}$, remains constant until $t = 1.0 \times 10^{-4} \text{ s}$, and is completely removed to 0 at $t = 1.5 \times 10^{-4} \text{ s}$, remaining zero thereafter. A constant time step of $\Delta t = 1.0 \times 10^{-5} \text{ s}$ is used for all simulations. For the discretization schemes, the spatial domain is uniformly discretized into a 40×4 mesh across all three methods. The results computed by the traditional FEM on this mesh serve as the exact reference solution. For both the traditional MPM and the IGA-MPM, this mesh acts as the background grid, and the elastic bar is represented by initially placing 25 material points per cell. We evaluate and compare the C^0 -continuous linear shape functions employed in the traditional MPM and the C^1 -continuous shape functions utilized in the IGA-MPM.

Figure 6.3.1a shows the comparison of the displacement u_x along the length of the bar at $t = 5.8 \times 10^{-4}$ s. It is evident that in predicting the displacement field, both the traditional MPM and IGA-MPM show excellent agreement with the FEM reference solution; the curves are very smooth, and no distinct errors are observed. However, the stress σ_{xx} distribution along the bar, shown in Figure 6.3.1b, vividly exposes the shortcomings of the traditional method. It can be seen that the case of C^0 -continuous linear shape functions (traditional MPM) produces severe non-physical oscillations, especially in the region between $X = 0.25$ m and 0.4 m, where drastic stress spikes occur and the curve as a whole exhibits a step-like pattern. This is a consequence of cell-crossing errors generated during particle movement, which are induced by the low-continuity grid shape functions. In contrast, in the IGA-MPM utilizing high-order continuous shape functions, these non-physical spikes and step-like phenomena are completely eliminated, yielding a smoothly varying stress field that highly coincides with the exact solution. This result strongly demonstrates the significant advantage of isogeometric basis functions in mitigating the inherent numerical noise and instability of the MPM.

6.3 Dynamic bending of an elastic cantilever beam

To further assess the performance of the proposed formulation in simulating dynamic responses involving large deformations, the dynamic bending of an elastic cantilever beam is investigated. The beam has a length of $L = 8$ m and a height of $H = 1$ m. A NEO-HOOKEAN hyperelastic material model is employed with a YOUNG's modulus of $E = 60$ Pa, a POISSON's ratio of $\nu = 0.3$, and a mass density of $\rho = 1 \times 10^{-6}$ kg/m³. The beam is fully clamped at its left boundary where $x = 0$ and is subjected to a transient dynamic excitation in the form of a downward pulse load. The load amplitude is linearly ramped up to its peak over 5×10^{-3} s, maintained until 1×10^{-2} s, and completely removed at 1.5×10^{-2} s. The total simulation time is set to $t = 1 \times 10^{-1}$ s with a constant time step of $\Delta t = 5 \times 10^{-4}$ s. For the spatial discretization of both the standard MPM with C^0 -continuous linear shape functions and the proposed IGA-MPM with C^1 -continuous quadratic shape functions, a background grid of 64×8 cells is employed, where 25 material points are uniformly distributed within each cell initially. Consistent with the previous examples, a high-resolution FEM simulation comprising 256×32 elements and employing the Newmark time integration scheme is utilized as the exact reference solution to benchmark the particle-based methods.

Figure 6.3.2 depicts the deformed configurations and the corresponding horizontal stress σ_{xx} contours for the three numerical methods. The results clearly expose the inherent vulnerabilities of the traditional MPM shown in Figure 6.3.2b under large dynamic deformations. Severe cell-crossing instabilities manifest as non-physical, checkerboard-like stress oscillations across the entire domain, particularly near the fixed boundary where the bending stresses are most concentrated. In stark contrast, the IGA-MPM depicted in Figure 6.3.2c effectively and completely suppresses these spatial oscillations. By leveraging the higher-order continuity of the B-spline basis functions, IGA-MPM yields a smooth, continuous stress field that is in excellent macroscopic and microscopic agreement with the reference FEM solution presented in Figure 6.3.2a.

To quantitatively evaluate the temporal accuracy and stability, the time histories of the horizontal stress σ_{xx} and the vertical displacement u_y are plotted in Figure 6.3.3. The stress is tracked at a material point with initial coordinates $x = 0.5625$ m and $y = 0.9375$ m located near the fixed support, while the displacement is recorded at a point with initial

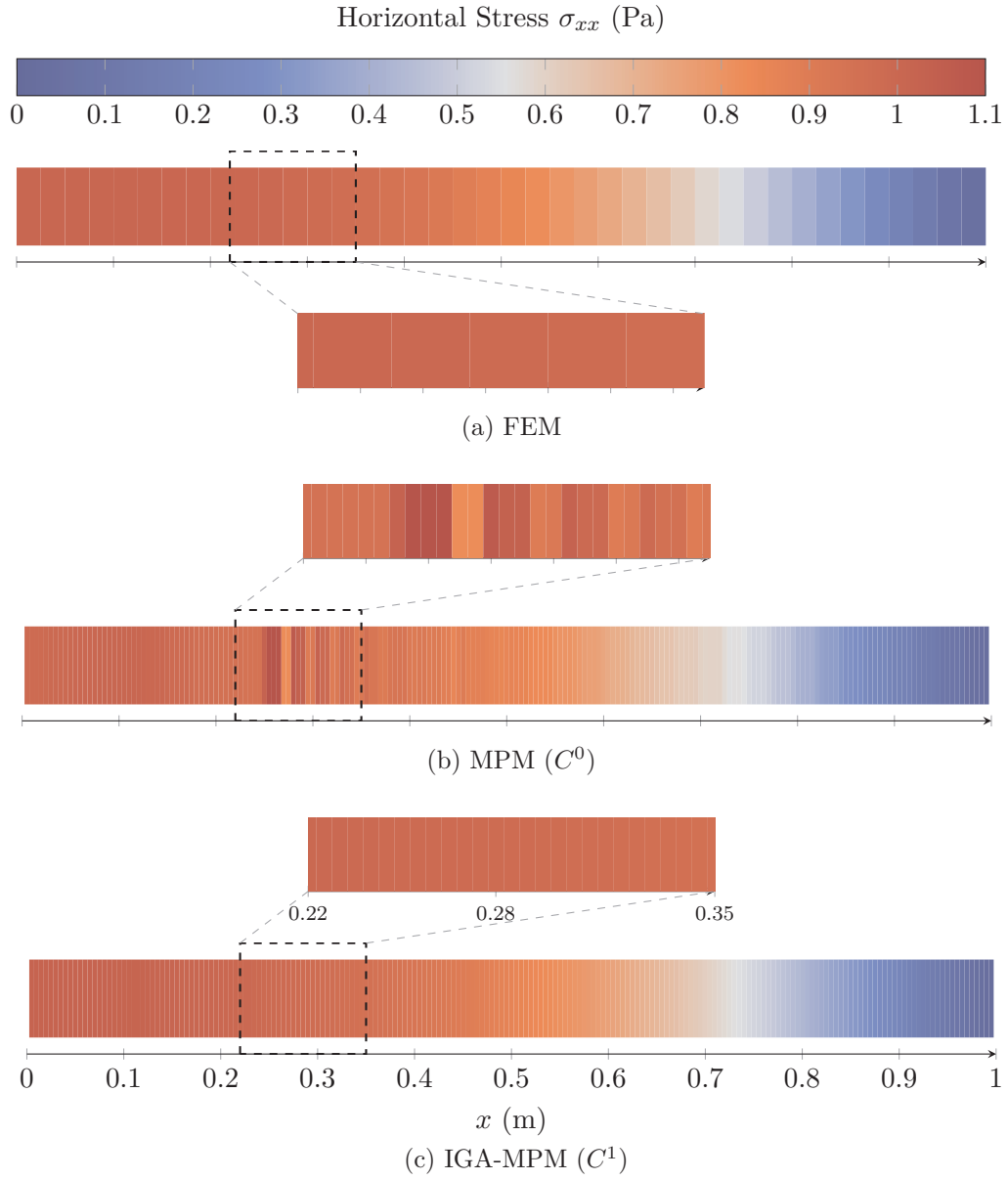
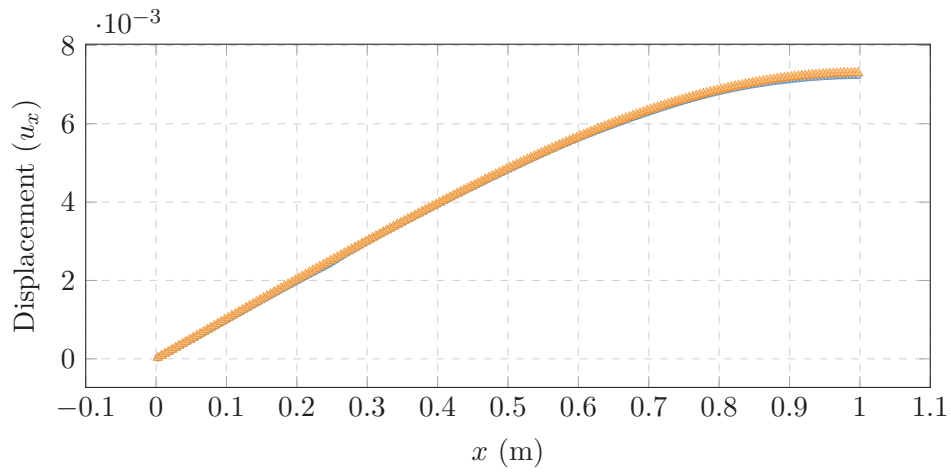
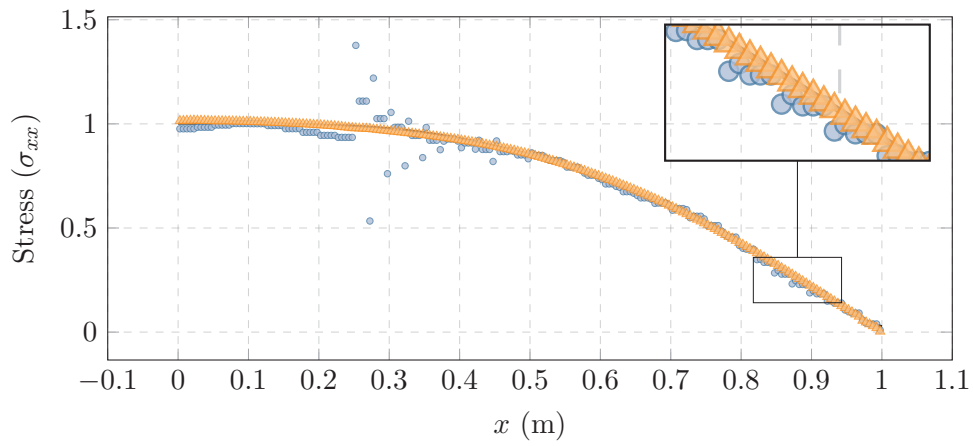


Figure 6.2.1: Comparison of horizontal stress $t = 5.8 \times 10^{-4}$ s.



(a) Displacement distribution



(b) Stress distribution



Figure 6.3.1: Comparison of displacement and stress among three methods.

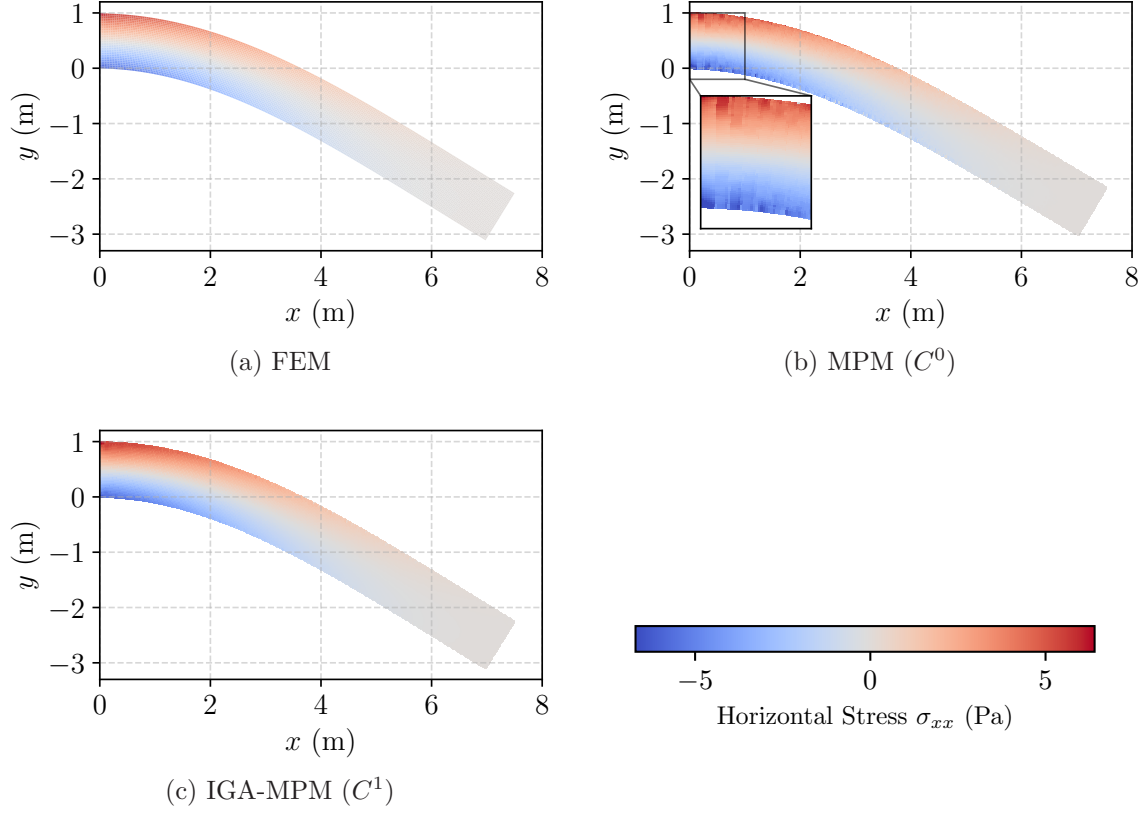


Figure 6.3.2: Comparison of horizontal stress contours among three methods at $t = 89$ step.

coordinates $x = 7.9375$ m and $y = 0.4375$ m near the free end. While the overall vertical displacement responses in Figure 6.3.3b appear globally similar across all formulations, the magnified view reveals that the IGA-MPM tracks the exact FEM reference curve much more precisely than the traditional MPM, avoiding slight deviations at the amplitude peaks. More importantly, the stress history illustrated in Figure 6.3.3a highlights the severe limitations of traditional MPM in dynamic scenarios. As material points repeatedly cross the boundaries of the background grid cells, the discontinuous gradients of the C^0 shape functions induce sharp, spurious high-frequency temporal oscillations in the stress response. The IGA-MPM, however, delivers a highly stable and smooth stress evolution, accurately capturing the dynamic bending behavior without introducing any noticeable numerical noise. This further demonstrates the superiority of the IGA-MPM in robustly evaluating stress-dependent variables in large-deformation dynamic problems.

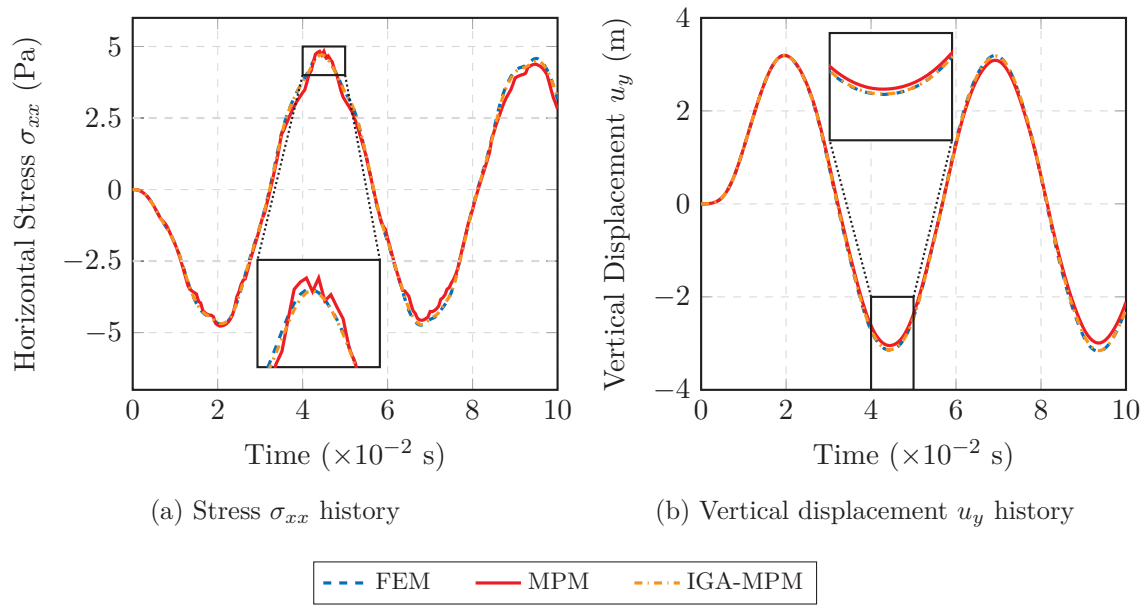


Figure 6.3.3: Comparison of horizontal stress and vertical displacement histories among three methods.

7 Conclusions

This thesis successfully extended the FEAP framework to improve the versatility and accuracy of MPM simulations. To address the inherent numerical instabilities of the traditional MPM, this work primarily accomplished four interconnected objectives. First, the B-spline MPM was implemented by leveraging the built-in IGA module within FEAP, substituting standard linear shape functions with higher-order continuous B-spline basis functions to fundamentally resolve the cell-crossing error. Second, support for transient MPM simulations was integrated into FEAP by extending its underlying dynamics code and the existing MPM architecture. Third, a mesh-based material point initialization algorithm was developed for standard MPM, which was subsequently optimized for IGA-MPM by utilizing smooth B-spline mapping to accurately seed particles and capture exact curved geometric boundaries without discretization gaps. Finally, the developed IGA-MPM was systematically evaluated and compared against the linear MPM and the traditional FEM through a series of representative benchmarks.

By evaluating both the traditional linear MPM and the proposed IGA-MPM within the same computational framework, a rigorous comparative analysis was conducted. The standard MPM utilizes linear C^0 -continuous shape functions, which cause severe cell-crossing errors and non-physical stress oscillations. The static cantilever bar analysis demonstrated that while the traditional MPM suffered from error plateaus due to these discontinuous gradients, the IGA-MPM achieved excellent spatial and quadrature convergence. Furthermore, the transient dynamics benchmarks—specifically a pulse impact on a 1D elastic bar and the dynamic bending of an elastic cantilever beam—verified that the higher-order continuity of B-splines intrinsically eliminates the step-like stress jumps and non-physical spatial oscillations under large deformations that plague the standard MPM. Overall, the developed framework successfully expands the application scope of the MPM and significantly improves its numerical accuracy.

8 Outlook

Building upon the established framework, several avenues for future research and enhancement can be identified:

- **Extension to Complex Material Models:** The current implementation utilizes a compressible Neo-Hookean hyperelastic model. Future work could integrate elastoplasticity, viscoplasticity, or damage mechanics to simulate complex material failure, ductile fracture, and severe fragmentation processes.
- **Contact and Impact Mechanics:** While the IGA-MPM provides smooth stress fields, implementing advanced contact algorithms would allow the framework to effectively simulate multi-body interactions, penetration, and high-velocity impact problems.
- **Multi-physics Coupling:** Extending the transient formulation to include thermo-mechanical coupling or fluid-structure interaction could broaden the application scope to complex multi-physics problems, such as high-temperature material forming or saturated soil dynamics.

Acronyms and Glossary

Acronyms

BEM	Boundary Element Method	17
CAD	Computer-Aided Design	8
CPDI	Convected Particle Domain Interpolation	30
FDM	Finite Difference Method	17
FEA	Finite Element Analysis	11
FEAP	Finite Element Analysis Program	1
FEM	Finite Element Method	1
FLIP	Fluid-Implicit-Particle	22
FVM	Finite Volume Method	17
G2P	Grid to Particle	22
GIMP	Generalized Interpolation Material Point	30
IGA	Isogeometric Analysis	1
IGA-MPM	Isogeometric Analysis Material Point Method	XI
MPM	Material Point Method	1
NURBS	Non-Uniform Rational B-Splines	1
P2G	Particle to Grid	25
PDE	Partial Differential Equation	17
PIC	Particle-in-Cell	22
PPC	Particles Per Cell	39
SPH	Smoothed Particle Hydrodynamics	21
TL	Total Lagrangian	26
UL	Updated Lagrangian	26

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